

An Agent-Based Model of Rumor-Induced Volatility in Financial Markets

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ABSTRACT

Rumors in financial markets impact investors' decisions, driving asset prices away from their fundamental valuations. From a regulatory perspective, it is challenging to contain such rumors. We develop an agent-based model to understand the price discovery process in a simulated stock market that allows heterogeneous agents, who differ in financial literacy and cognition to interact for price formation. We show that both financial literacy and cognition are important determinants of rumor spread in stock markets: Higher (lower) cognition and higher (lower) financial literacy reduce (increase) rumor spread. Our results suggest that both the prevalence and intensity of financial literacy play a significant role in reducing rumor induced volatility and promoting market stability.

1. Introduction

Regulators around the world are actively engaging in combating financial misinformation and rumors. Despite the efforts of regulators, investors often trade based on investment advice found online (Ammann and Schaub, 2021). The rise of social media has amplified the reach of rumors by financial influencers and has made it difficult for investors to make informed investment decisions. While social media is a source of collective wisdom (Chen et al., 2014; Da and Huang, 2020), it can also amplify herding, overreaction (Ahern and Peress, 2023) and asset price volatility (Shiller, 2014). Existing studies on rumors have been largely theoretical, examining them as a form of strategic communication where informed investors spread rumors to followers to gain profits (Van Bommel, 2003), with credibility being influenced by their investment horizon (Schmidt and Goldstein, 2020).

Models of herding and information cascades consider the decisions of investors to critically depend on peers (Banerjee, 1993, 1992; Bikhchandani et al., 1998, 1992; Bikhchandani and Sharma, 2001). Social interactions and peer influence can play a significant role in shaping investor behavior (Ozsoylev et al., 2013; Pedersen, 2022). Apart from social influence, recent studies suggest that investor specific attributes shape investor behavior and in turn can influence asset prices (Han et al., 2022; Hirshleifer and Teoh, 2008). When investors analyze each investment option, their decisions can be driven more by their endogenous traits rather than by the behavior of their peers (Akçay and Hirshleifer, 2020). Psychological biases stemming from investors' individual traits can systematically influence decision making under uncertainty (Daniel et al., 2002; Barberis and Thaler, 2003; Hirshleifer, 2001; Shiller, 2003). Apart from risk aversion, investors' psychology can affect willingness to take risks (Kahneman and Tversky, 1979; Tversky and Kahneman, 1974). Rational decision making may be constrained by cognitive limitations, time and information availability as suggested by the concept of bounded rationality of investors (Simon, 1955; Arthur, 1991).

Several studies show that cognition shapes investors' risk perception and decision-making processes. Cognition is a broad term covering attention, thinking, memory, learning and perception. Several studies have considered IQ as a proxy for the cognition of an individual (D'Acunto et al., 2023; Dohmen et al., 2018; Grinblatt et al., 2011). While some investors may be impulsive and follow a rumor in anticipation of large gains, others may engage in more deliberate decision-making¹. Investors with high cognitive ability tend to be more patient (Agarwal and Mazumder, 2013; Benjamin et al., 2013; Dohmen et al., 2010; Frederick, 2005; Oechssler et al., 2009) and are more likely to choose delayed rewards over immediate gains (Mischel and Shoda, 1995). Their ability to discern when risk taking is advantageous can enhance investment decisions (Dohmen et al., 2018; Grinblatt et al., 2011). Thus, risk preference may be shaped by cognitive abilities. Investors with low cognition tend to exhibit greater overconfidence and take on more risk (Christelis et al., 2010). (Christelis et al., 2010). This can be detrimental during rumors.

Financial decisions of investors can also critically depend on their financial literacy (Lusardi and Mitchell, 2014). Financial literacy can reduce individuals' cognitive biases (IOSCO and OECD, 2018) and transform financial behavior

ORCID(s):

¹Two system theories state that a person can exhibit long run patient behavior and impulsive short run behavior.

(Kaiser et al., 2022; Kaiser and Menkhoff, 2017). Individuals with high financial literacy will seek more formal channels of advice (financial advisors) and rely less on word-of-mouth communication (Calcagno and Monticone, 2015). Higher cognition can enhance financial literacy by making learning processes easier (Muñoz-Murillo et al., 2020). However, even investors with high cognition may be prone to biases (Benjamin et al., 2013; Oechssler et al., 2009). The efforts to increase financial literacy may be limited by cognitive biases (García, 2013). When faced with the same information set investors can differ in their information processing capabilities leading to heterogeneous beliefs about the assets payoffs (Lux, 1998; Anderson et al., 2005).

Thus, investors can differ in their ability to process information during rumors depending on their cognition and financial literacy which can influence their decisions under risk and uncertainty. While high financial literacy or cognition alone may not guarantee rational decision-making during rumors, sophisticated investors who have both high cognition and financial literacy can be resilient to rumors. To study the role of financial literacy and cognition of investors on asset price dynamics, a modeling paradigm is necessary that can link these attributes to asset prices.

Agent Based Models (ABMs) have explained empirical phenomena in financial markets with the help of heterogeneous agents following unique trading strategies (Axtell and Farmer, 2025; Hommes, 2006; Hommes and Wagener, 2008). Rumors can trigger volatility in asset prices, yet the mechanisms that drive volatility remain understudied. To this end, we contribute a novel agent-based model of rumor dynamics in financial markets that endogenizes investor heterogeneity in both financial literacy and cognitive ability, and links these traits to volatility and price outcomes under information shocks.

Agent-based modeling (ABM) is a modeling technique that allows observing and optimizing strategic interactions between autonomous, goal-oriented decision-making entities. Existing ABMs have shown that heterogeneous beliefs can lead to upward (downward) markets (Day and Huang, 1990), bubbles and crashes (Lux, 1995; Westphal and Sornette, 2023, 2020) and complex dynamics of asset prices (Chiarella, 1992). While heterogeneity in the form of fundamentalists and chartists has been considered, heterogeneity within individuals has not been considered explicitly. Existing ABMs often treat individual agents as uniform internally. This overlooks cognitive and informational asymmetries such as differing financial literacy and decision thresholds that can affect behavior during rumors. Unlike traditional ABMs that assume behaviorally homogeneous types, we endogenize susceptibility to misinformation through dual internal traits - financial literacy and cognition which jointly determine agent transitions and demand functions. This represents a behavioral deepening of the existing frameworks.

Through simulation experiments we find that financial literacy and cognition both influence the scale and spread of rumors. Lower financial literacy and cognition lead to higher number of agents spreading the rumor. When the percentage of agents with high financial literacy and cognition increases beyond 30%, the maximum return and volatility of asset prices observed decreases. The time taken for rumor to dissipate also decreases as the percentage of investors with high financial literacy and cognition increases. The rest of the paper is organized as follows: Section 2 presents the design and working of the ABM, Section 3 discusses the asset price formation, Section 4 presents the results, sensitivity analysis of varying parameter values and the validation of outputs. Section 5 presents the discussion of results and Section 6 discusses the policy implications of this study.

2. The Design of the ABM

To study the role of financial literacy and IQ of investors on asset price dynamics, we develop an agent-based model (ABM) in *NetLogo*, a popular multi-agent programming environment. In the model, investors on a social media network are considered 'agents'. The total number of agents is determined by *numberOfInvestors* parameter. All the agents are randomly distributed across patches. An agent i can be in one of the following states at time t :

- **Spreader** (N_t): Agents in this state actively spread the rumor.
- **Exposed** (E_t): Agents in this state are exposed to the rumor but do not spread it.
- **Susceptible** (S_t): Agents in this state are connected to the agents in the spreader or exposed state.
- **Neutral** (I_t): Agents in this state do not follow or react to the rumor.

At each discrete time step t , an agent i is characterized by the following attributes:

- **Financial literacy** $f_t^i \in [0, 100]$ — A measure of agents' i 's financial literacy at time t

- **IQ** $iq_t^i \in [50, 150]$ — Cognitive ability of an agent i
- **Wealth** W_t^i — Total wealth (cash + shares) of agent i at time t
- **State variable** S_t (Susceptible), E_t (Exposed), N_t (Spreader), I_t (Neutral)
- **Sick-time** t_s — Number of time steps an agent remains in a state other than the neutral state.
- **Base followers** — The initial number of followers of an agent.

The investor state transition is explained in Figure 1.

- Spreaders post rumors to their followers. Followers of agents in spreader state transition to susceptible state.
- If an agent remains in the susceptible state for some time, they transition to exposed state:

$$\text{if } t_s > 3 \text{ and } S_t \text{ then: } \begin{cases} E_t & \text{if } f_t^i > \text{random}(100) \text{ and } iq_t^i > \text{random}(100) \end{cases}$$

$$\text{if } t_s > 10 \text{ and } E_t \text{ then: } \begin{cases} I_t & \text{if } f_t^i + iq_t^i > \text{literacyThreshold} + iqThreshold \\ N_t & \text{otherwise} \end{cases}$$

All agents are in the neutral state at the beginning of the simulation. A certain percentage of the total number of agents controlled by *initialSpreaders* parameter are initialized in the spreader state. The followers of agents in spreader state become susceptible. Uncertainty may exist regarding the threshold at which financial literacy is considered high. To determine whether financial literacy is high or not we design a parameter *literacyThreshold*. Agents having financial literacy above this threshold are considered highly financially literate. The percentage of agents with high financial literacy is determined by *percentHighLiteracy* parameter. These agents have financial literacy in the range [*literacyThreshold*,100]. Similarly to determine whether IQ is high or not we design a parameter *iqThreshold*. The percentage of agents with high IQ is determined by *percentHighIQ* parameter. These agents have IQ in the range [*iqThreshold*,150]. The range of IQ is taken to be in accordance with WAIS (Wechsler Adult Intelligence Scale) which is widely used scale to measure IQ. The number of followers for each agent *baseFollowers* is initialized as a random draw from a normal distribution with mean *averageConnections* and standard deviation *stdDevConnections*. Each agent is connected to their followers through a wiring algorithm. At each time step t , the deviation in the number of followers is determined by *followersDeviation* parameter. Agents follow or unfollow each other forming dynamic connections with other agents. If the agents in the spreader state post or share the rumor they receive, their followers who are in the susceptible state transition to the exposed state at random probability. The exposed agents then assess the rumor based on their financial literacy and IQ. If the sum of financial literacy and IQ is greater than or equal to the sum of *literacyThreshold* and *iqThreshold*, then exposed agents transition to neutral state otherwise they transition to the spreader state. Higher cognition and financial literacy can enhance resistance to misinformation. More complex transition functions could be employed such as a weighted average of cognition and financial literacy or a non linear transition function where higher cognition can lead to a more pronounced gain in financial literacy. However to consider both traits as equally important we consider a simple summation.

The financial literacy of agents in spreader state is incremented by *literacyAdder* parameter. The educate parameter is modeled to replicate a regulator actively disseminating awareness. If the sum of financial literacy and IQ exceed the sum of *literacyThreshold* and *iqThreshold* then agents in the spreader state transition to the neutral state, otherwise they remain in the spreader state. Agents in spreader state actively spread the rumor to their followers. They buy(sell) the asset during a positive(negative) rumor. Agents in the neutral state assess whether their followers are spreading the rumor. They unfollow agents actively spreading the rumor thereby reducing their exposure to misinformation. Agents in neutral state buy(sell) when the asset is undervalued(overvalued). In the model, the asset price is driven by endogenous changes in the fraction of agents in the spreader and the neutral state. We model the demand for the asset from agents in spreader state as a function of their combined cognitive ability, which reflects the influence of both financial literacy and IQ. The asset price formation is discussed in detail in the next section. The emergent features of the model are the number of agents who are in the susceptible, exposed, spreader and neutral state and the asset price, returns and volatility. All the model parameters are summarized in Table 1.

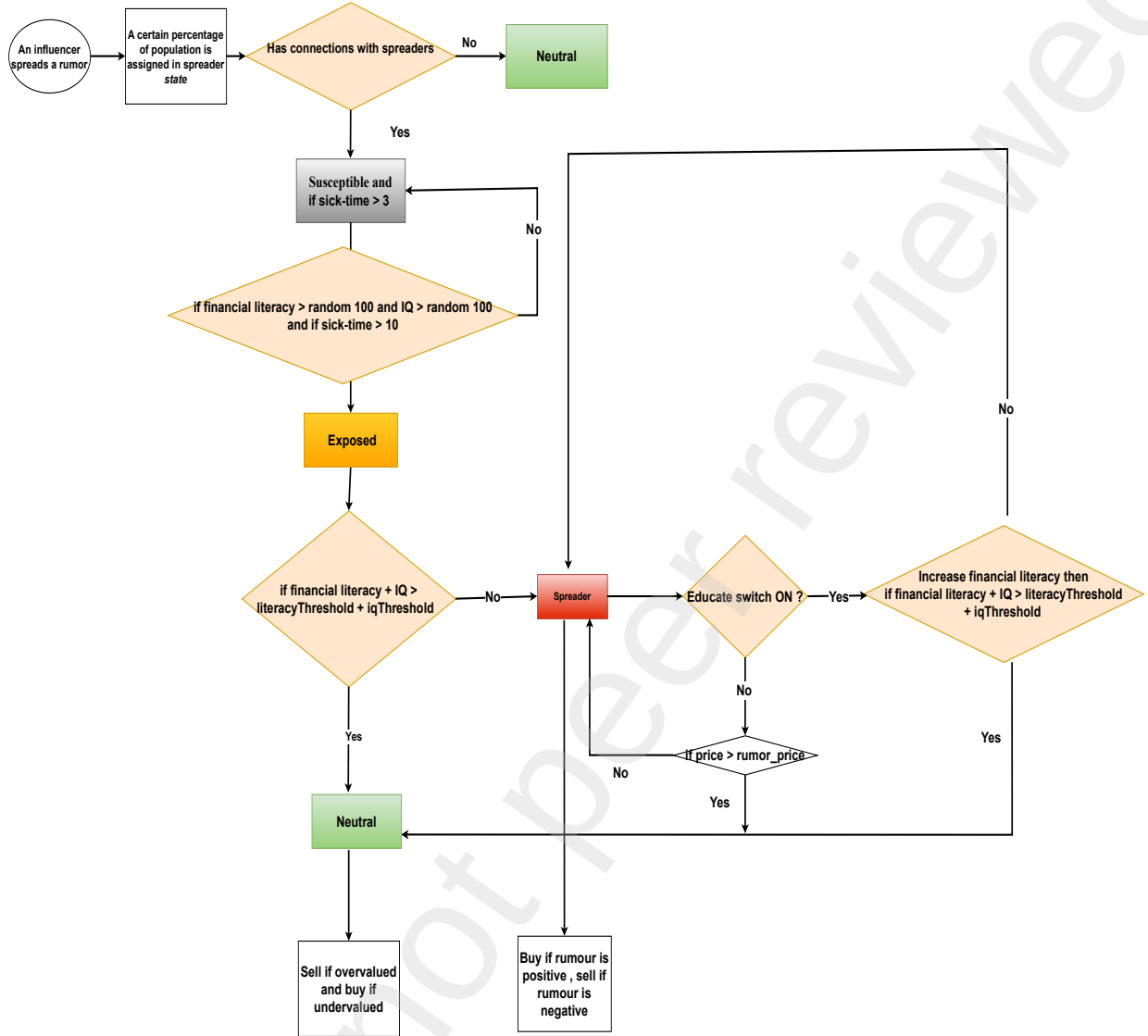


Figure 1: Investor state transition diagram.

3. Asset Demand and Price Formation

Agents in the spreader state buy (or sell) assets based on the rumor. Changes in the number of spreaders and neutral agents affects the net demand for the asset. We modify the Franke and Westerhoff model for asset price formation (Franke and Westerhoff, 2012; Pruna et al., 2018) to incorporate a cognitive and behavioral aspect of demand from spreaders. Spreaders expect price to converge to the rumor price and act based on anchored expectations as opposed to chartists who base their expectations on recent price changes. Agents in the neutral state base their demand on the mispricing of the asset relative to the fundamental price. The price of the asset at time t is denoted by p_t , the rumor price is denoted by p_t^r , and the fundamental price is denoted by p_t^f . The demand from spreaders is formulated as a function of the deviation between the anticipated rumor price and the market price, i.e., $(p_t^r - p_{t-1})$. The demand from neutral agents is proportional to the difference $(p_t^f - p_t)$. The value of demand for the asset at time period t from each agent in neutral state is given by:

Table 1

Social network parameters: descriptions, ranges, and values

Parameter	Description	Range	Value
<i>numberOfInvestors</i>	The number of agents in the social network	1–1000	600
<i>averageConnections</i>	Average number of connections per agent	1–40	27
<i>stdDeviationConnections</i>	Standard deviation in the number of connections	1–30	6
<i>literacyAdder</i>	The value by which financial literacy is increased	1–5	5
<i>iqThreshold</i>	The value above which IQ of agents is considered high	90–140	120
<i>literacyThreshold</i>	The value above which financial literacy is considered high	50–90	80
<i>percentHighLiteracy</i>	The percentage of agents with high financial literacy	10–100	Varied
<i>percentHighIQ</i>	The percentage of agents with high IQ	10–100	Varied
<i>initialSpreaders</i>	The percentage of agents in the spreader state at the beginning	10–60	10
<i>followersDeviation</i>	Standard deviation in the number of followers	0–6	2

$$d_t^f = \phi \cdot (p_t^f - p_t) + \varepsilon_t^f, \quad \varepsilon_t^f \sim \mathcal{N}(0, \sigma_f^2), \quad \phi_i \in (0, 1)$$

The value of demand for the asset at time period t from an agent i in spreader state is given by:

$$d_t^s = \psi_i \cdot (p_t^r - p_{t-1}) + \varepsilon_t^s, \quad \varepsilon_t^s \sim \mathcal{N}(0, \sigma_s^2)$$

If the rumor is negative, then:

$$d_t^s = \psi_i \cdot (p_{t-1} - p_t^r) + \varepsilon_t^s, \quad \varepsilon_t^s \sim \mathcal{N}(0, \sigma_s^2)$$

Here, ϕ is a constant denoting aggressiveness of trading of an agent in the neutral state and ψ_i is the aggressiveness of trading of an agent i in spreader state. It is given by:

$$\psi_i = 1 - \frac{1}{1 + \exp(-a \cdot (c_i - b))}, \quad \psi_i \in (0, 1)$$

$$c_i = 0.5 \cdot f_t^i + 0.5 \cdot iq_t^i$$

where f_t^i and iq_t^i denote the financial literacy and IQ of agent i at time t , respectively. f_t^i is capped at a maximum value of 100. The sigmoid function transforms the combined score c_i . The steepness of the sigmoid curve is determined by $a \in (0, 1)$, and its center by $b \in [75, 125]$. The minimum and maximum possible values of the sum of f_t^i and iq_t^i are 50 and 150; as a result, $c_i \in [25, 125]$.

Here, the demand coefficient ψ_i is highest when both f_t^i and iq_t^i are at their minimum. This coefficient represents the order size or the number of shares demanded. When agents have a high combined cognitive score c_i , their demand for the asset is minimal. The rumor price evolves over time via exponential decay:

$$p_{t+1}^r = (1 - \delta) \cdot p_t^r + \delta \cdot p_t$$

where $\delta \in [0, 1]$ is the rumor decay parameter.

During a positive (negative) rumor, the sign of demand from spreaders is +1 (−1), and the demand from the neutral agents is positive (negative) if the asset is undervalued (overvalued).

The equation determining the price in the next period t is given by:

$$p_{t+1} = p_t + \mu \cdot (d_t^f \cdot n_t^f + d_t^s \cdot n_t^s) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_t^2)$$

where

$$n_t^f = \frac{\text{number of agents in neutral state } (I_t) \text{ at time } t}{\text{total number of agents}},$$

$$n_t^s = \frac{\text{number of agents in spreader } (N_t) \text{ state at time } t}{\text{total number of agents}},$$

and $\mu > 0$ is the price adjustment coefficient.

Table 2
Parameters related to asset price

Parameter	Description	Value
p_t	Initial price of the asset	336
p_f	Fundamental price	300
ϕ	Aggressiveness of neutral agents	0.5
σ_f^2	Noise variance in fundamental demand	0.01
σ_s^2	Noise variance in spreader demand	0.01
σ_t^2	Variance in asset price	0.01
δ	Rumor decay parameter	0.01
μ	Price adjustment coefficient	0.12
a	Component of ψ	0.05
b	Component of ψ	125

The rumor price is taken as 600 for a positive rumor and 200 for a negative rumor. The values of the parameters ϕ , σ_f^2 , σ_s^2 , σ_t^2 , and μ are calibrated using the method of moments based on the past 5 years of daily stock price data for Tesla. The method of simulated moments (MSM) has been utilized to estimate the parameters of ABMs (Franke, 2009; Franke and Westerhoff, 2012). These values are reported in Table 2. The calibration follows a moment-matching approach, in which parameters are optimized so that key statistical moments of the simulated return distribution match those of the empirical return distribution. The following empirical moments are computed and used as targets for calibration: mean return, standard deviation of returns, first-order autocorrelation of returns and kurtosis of returns. The loss function penalizes squared deviations between simulated and empirical moments.

$$\text{Loss} = (\bar{r}_{\text{sim}} - \bar{r}_{\text{emp}})^2 + (\sigma_{\text{sim}} - \sigma_{\text{emp}})^2 + (\rho_{\text{sim}} - \rho_{\text{emp}})^2 + (\kappa_{\text{sim}} - \kappa_{\text{emp}})^2$$

where $\bar{r}$, σ , ρ , and κ represent the mean, standard deviation, first-order autocorrelation, and kurtosis of returns, respectively. Parameter optimization is performed using the Nelder-Mead method (Chang, 2012; Nelder and Mead, 1965). The optimization is initialized with the guess [0.5, 0.5, 0.01, 0.01, 0.1, 0.01] and constrained to non-negative values. The number of shares, value of shares, cash, and total wealth of each agent is recalculated and updated at each period t . In the case of demand to buy(sell) the asset, the value of shares held by an agent changes as:

$$\text{shares_value} = \text{shares_value} \pm \text{shares_value_transacted}$$

where $\text{shares_value_transacted}$ is the value of demand from each agent type. The number of shares is then updated as:

$$\text{shares} = \frac{\text{shares_value}}{p_{t-1}}$$

Correspondingly, the cash held by the agent is updated as:

$$\text{cash} = \text{cash} \mp \text{shares_value_transacted}$$

The total wealth is calculated as:

$$\text{wealth} = \text{shares} \cdot p_{t-1} + \text{cash}$$

As the rumor spreads, the number of agents in the spreader state increases initially, while the number of agents in the neutral state decreases. As the financial literacy of agents increases, the number of spreaders declines and the number of neutral agents rises. The simulation is terminated when the number of spreaders reaches zero.

4. Results

To investigate the influence of financial literacy and IQ on the scale and spread of rumor and their effects on asset price, return, and volatility, we conduct a set of experiments. In the first experiment, we set the rumor type to "positive" and set the *educate* parameter ON. We vary the percentage of agents with high financial literacy from 10% to 100% in steps of 10, while holding the percentage of agents with high IQ constant at 10%. The purpose of this experiment is to evaluate the effect of increasing financial literacy while holding cognition of agents constant. All other parameter values are the same as those listed in Table 1. In the second experiment, we vary IQ, while holding the percentage of agents with high financial literacy constant at 10%, with all other parameters the same as in the first experiment. To test whether the observed effects of financial literacy and IQ on rumor diffusion and asset price dynamics are robust across different rumor types (positive vs. negative), we repeat the first experiment with the rumor type changed to "negative" where percentage of agents with high financial literacy is increased while holding the percentage of agents with high IQ constant.

The results of the first experiment are presented in Figure 2. As the percentage of agents with high financial literacy increases, the time for the rumor to dissipate decreases, as shown in Figure 2(a). The scale of rumor spread is measured by two metrics: (i) the maximum number of agents in the spreader state and (ii) the minimum number of agents in the neutral state at any point in the simulation. As observed in Figure 2(b) and Figure 2(c), the maximum number of spreaders decreases and the minimum number of neutral agents increases with increase in percentage of agents with high financial literacy. When only 10% of agents have high financial literacy, the asset price reaches as high as 510. However, when the market is dominated by financially literate agents, the average asset price remains closer to the initial price of 336. Furthermore, both the maximum return and volatility observed in the market decrease significantly when the percentage of agents with high financial literacy exceeds 30%. The results of the experiment when percentage of agents with high IQ is varied and percentage of agents with high financial literacy is set at 10% are illustrated in Figure 3. We observe that as the percentage of agents with high IQ increases time for rumor to dissipate decreases. Maximum price observed, maximum return and volatility decrease steadily when percentage of population with high IQ increases beyond 30%.

The results of the experiment when percentage of agents with high IQ is varied and percentage of agents with high financial literacy is set at 10% are illustrated in Figure 3. We observe that as percentage of agents with high IQ increases time for rumor to dissipate decreases. Maximum price observed, maximum return and volatility decrease steadily when percentage of agents with high IQ increases beyond 30%. The results suggest that higher cognition may compensate for lower financial literacy, highlighting a complementary relationship between the two attributes in the spread of rumors and asset price dynamics.

Figure 4 reports the results when rumor is "negative" and percentage of agents with high financial literacy is varied and percentage of agents with high IQ is fixed at 10%. The minimum price observed increases from 250 to 325 as the percentage of agents with high financial literacy increases from 10% to 100%. This suggests that during a negative rumor the asset prices do not decline extremely when high literacy agents dominate. Large negative returns of about 3.2% are observed when percentage of agents with high financial literacy is low at 10%, 20% and 30%.

4.1. Sensitivity analysis

To study the effect of social network parameters on the results of the model, we perform systematic sensitivity analysis by varying one parameter at a time. The sensitivity analysis of parameters when the percentage of agents with high financial literacy is varied and percentage of agents with high IQ is kept constant at 10% for a positive rumor

is reported in the Appendix. The sensitivity analysis reveals that the `literacyAdder` parameter has a significant impact on the simulation outcomes. A decrease in the value of literacy added leads to a notable increase in the maximum observed values of price, return, and volatility. This suggests that increasing financial literacy of agents has significant impact on asset price dynamics during rumors. When the amount of literacy added is low (`literacyAdder` = 1), it leads to a higher number of agents in the spreader state. Also, the minimum number of agents in the neutral state decreases with decrease in the value of financial literacy added. This parameter has an effect on maximum price observed. The asset price deviates significantly from the initial price of 336 and reaches 625 when the percentage of agents with high financial literacy is 10% and value of literacy added is 1. When the value of literacy added is increased from 1 to 5, the maximum price observed decreases from 625 to 510 when the percentage of agents with high financial literacy is 10%. The value of maximum price observed lie between 620 to 550 when the value of literacy added is 1 suggesting that low value of literacy added leads to the price reaching the rumor price. The maximum return and volatility observed is also higher when the value of literacy added is low. When the percentage of agents with high financial literacy is 100%, the maximum volatility decreases from 2.2 to 1.6 with increase in literacy added from 1 to 5 as observed in Figure 5(e).

4.2. Output Validation

We validate our model by comparing the simulated return series against well established empirical regularities known as stylized facts. (Cont, 2001; Chakraborti et al., 2011; Lux and Marchesi, 1998). Specifically we validate three key properties of asset returns (i) absence of autocorrelation in raw return series, (ii) heavy tails in distribution of returns and (iii) volatility clustering. These properties serve as benchmarks to evaluate empirical plausibility of agent-based financial market models. Our model demonstrates face validity, as the simulated return series qualitatively mirror the patterns observed in real markets (Llacay and Peffer, 2018). The autocorrelation function ACF of returns is small and within the 95% confidence interval, suggesting lack of autocorrelation in return series as observed in the Figure 6. The QQ plot shows slight deviation from normality and the histogram of returns in Figure 6 (d) shows the distribution of return series is more leptokurtic than the normal distribution as observed in real markets (LeBaron et al., 1999). Leptokurtic returns have been observed due to herding by investors (Alfarano et al., 2005).

5. Discussion of Results

Our findings suggest that financial literacy and IQ of agents are important attributes that significantly influence the scale and spread of rumors. When the population either consists predominantly of agents with high financial literacy or high cognition asset price volatility decreases. Adding financial literacy has notable effect on the asset price. Asset prices do not deviate largely from the initial values with higher value of literacy added. In this study we modeled the spread of rumors considering simple sum of both the attribute values to affect agent transitions during rumors. More complex and non linear modeling rules could also be used such as bayesian or reinforcement learning mechanisms where agents update their beliefs over time based on personal traits or decisions of peers allowing dynamic adjustment in rumor susceptibility. Our model captures key dynamics of rumor propagation and market impact while also being tractable. Effect of varying network structures on rumor dynamics can be considered in future. Further research could improve realism by including real social media rumor data and consider more complex belief updating and learning mechanisms of agents.

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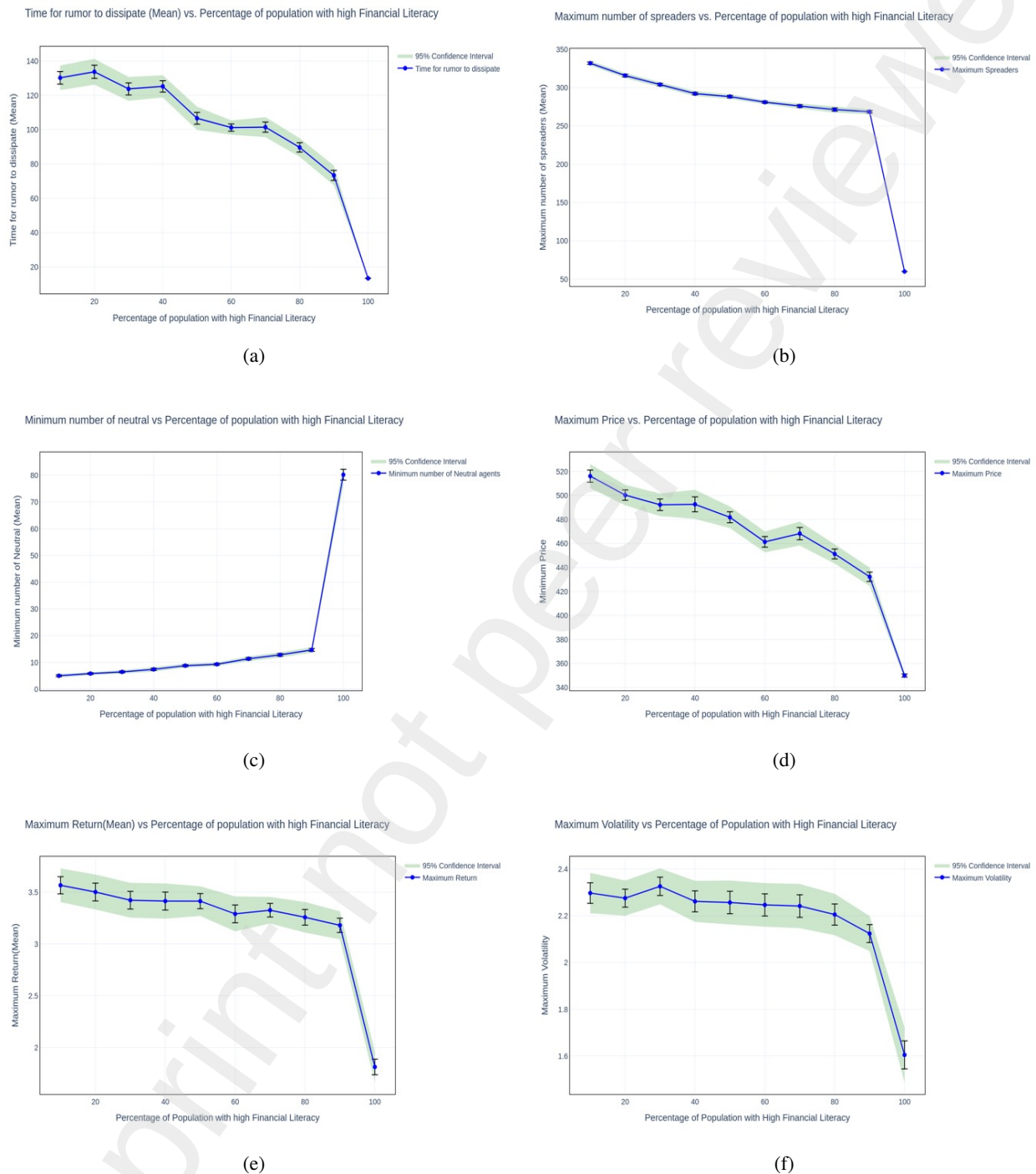


Figure 2: Percentage of agents with high financial literacy vs time for rumor to dissipate (a), maximum agents in spreader state at any point in the simulation (b), minimum number of agents in neutral state (c), maximum price (d), maximum return (e), and maximum volatility (f) across 50 simulation replications. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate.

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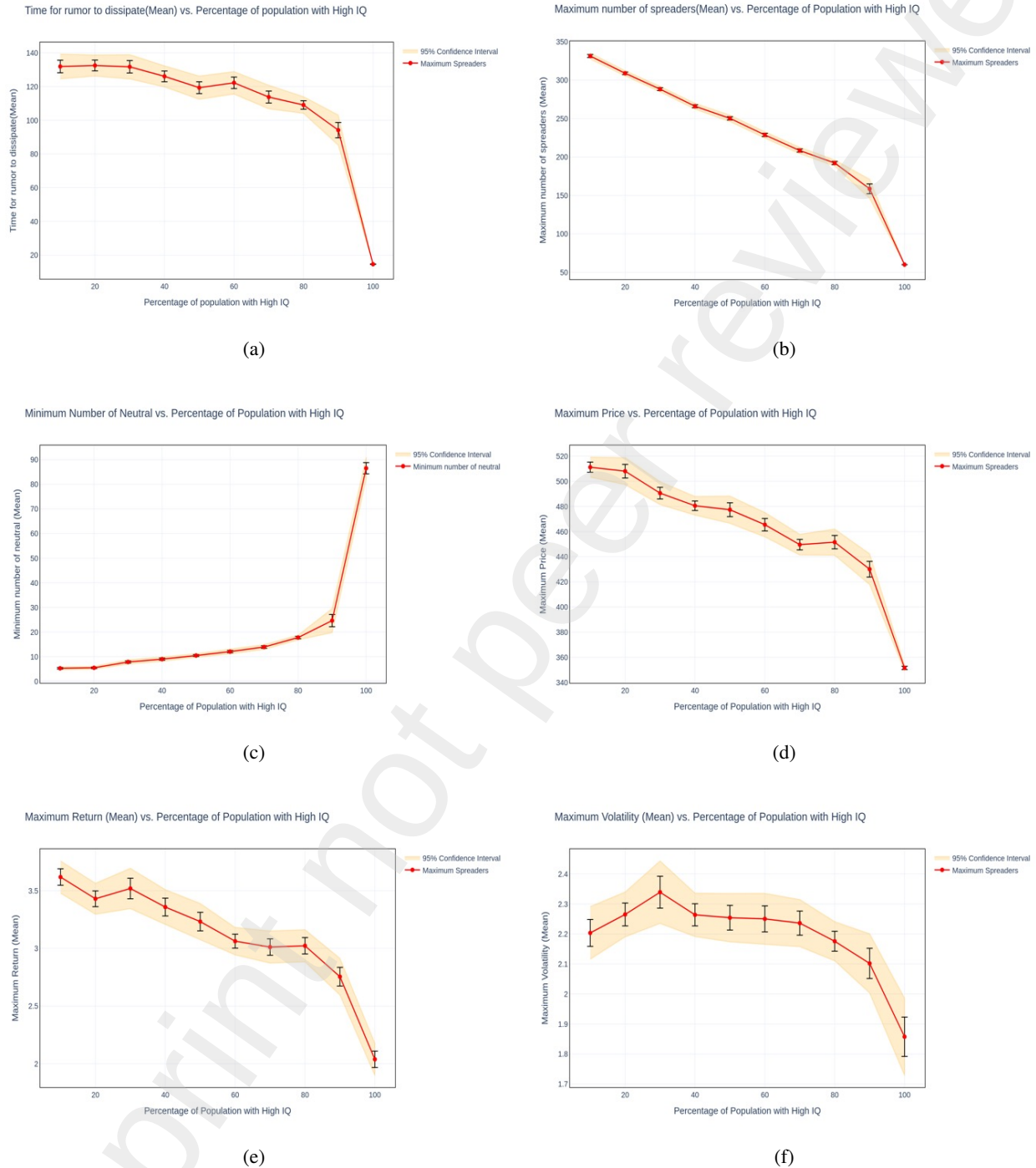


Figure 3: Percentage of agents with high IQ vs time for rumor to dissipate (a), maximum agents in spreader state at any point in the simulation (b), minimum number of agents in neutral state (c), maximum price (d), maximum return (e), and maximum volatility (f) across 50 simulation replications. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate.

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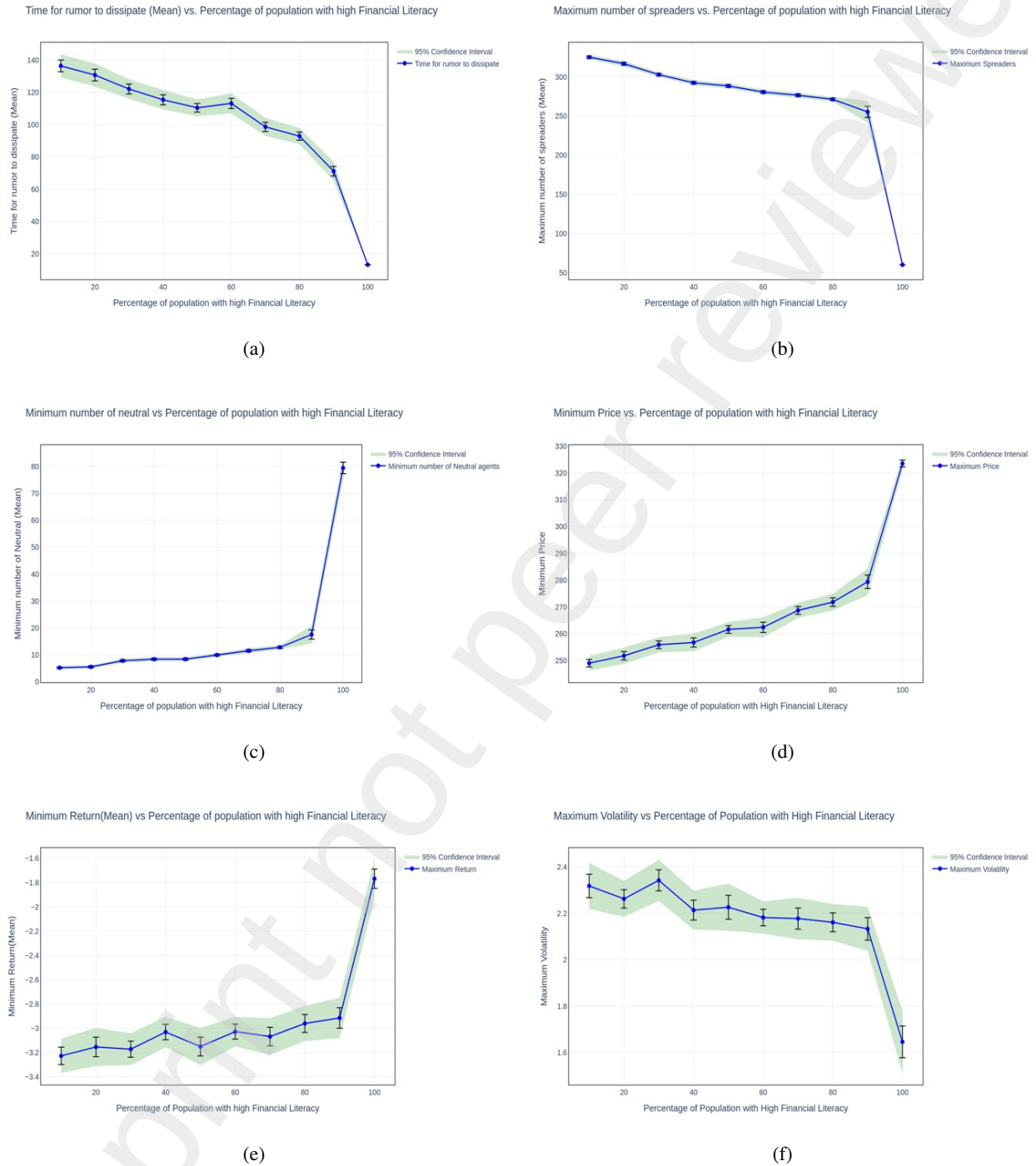


Figure 4: Percentage of agents with high financial literacy vs time for rumor to dissipate (a), maximum agents in spreader state at any point in the simulation (b), minimum number of agents in the neutral state (c), minimum price (d), maximum return (e), and maximum volatility (f) across 50 simulation replications for a negative rumor. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate.

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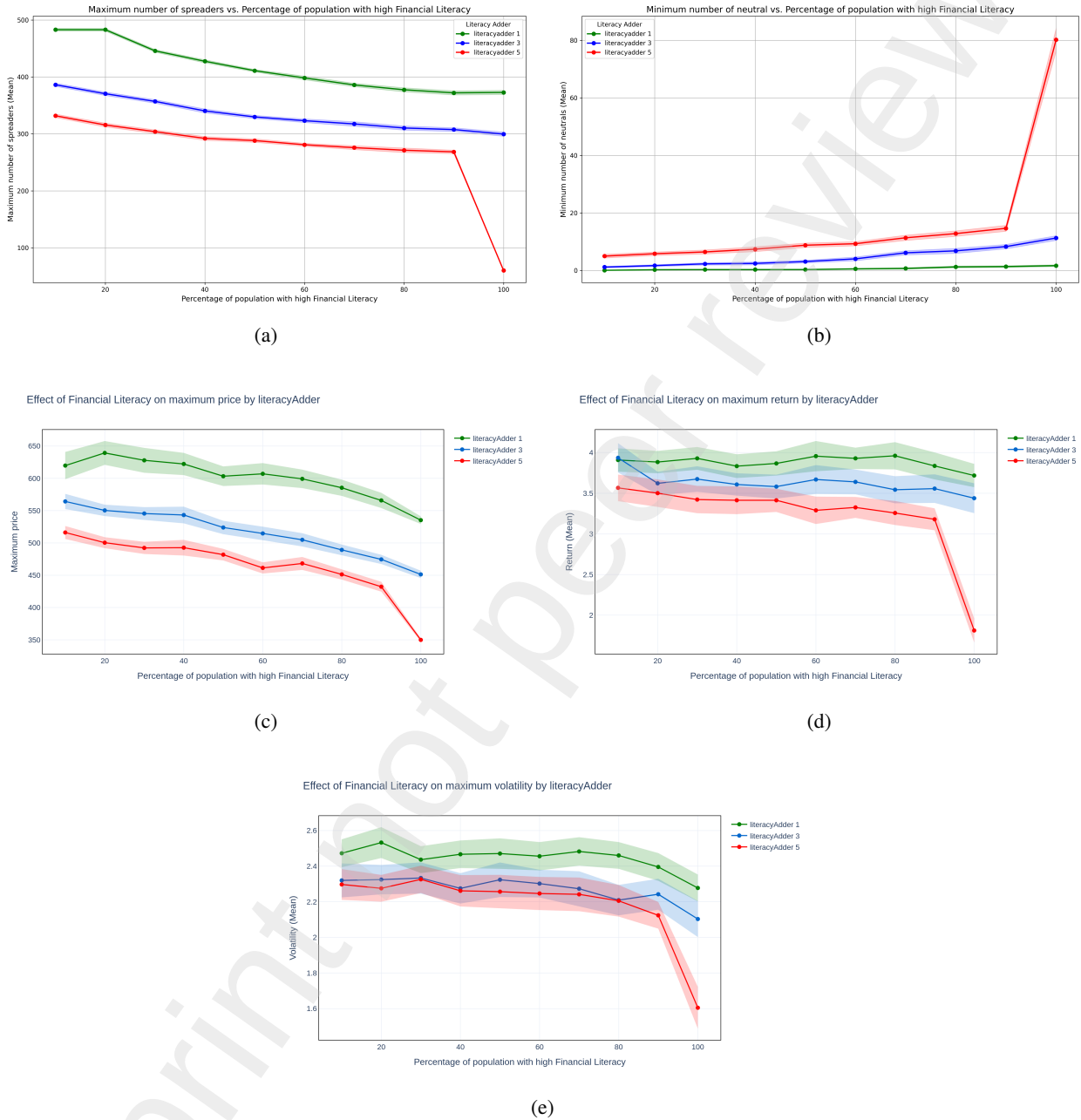


Figure 5: Effect of different literacyAdder on maximum number of spreaders (a), minimum number of neutral (b), maximum price (c), maximum return (d), and maximum volatility (e) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate.

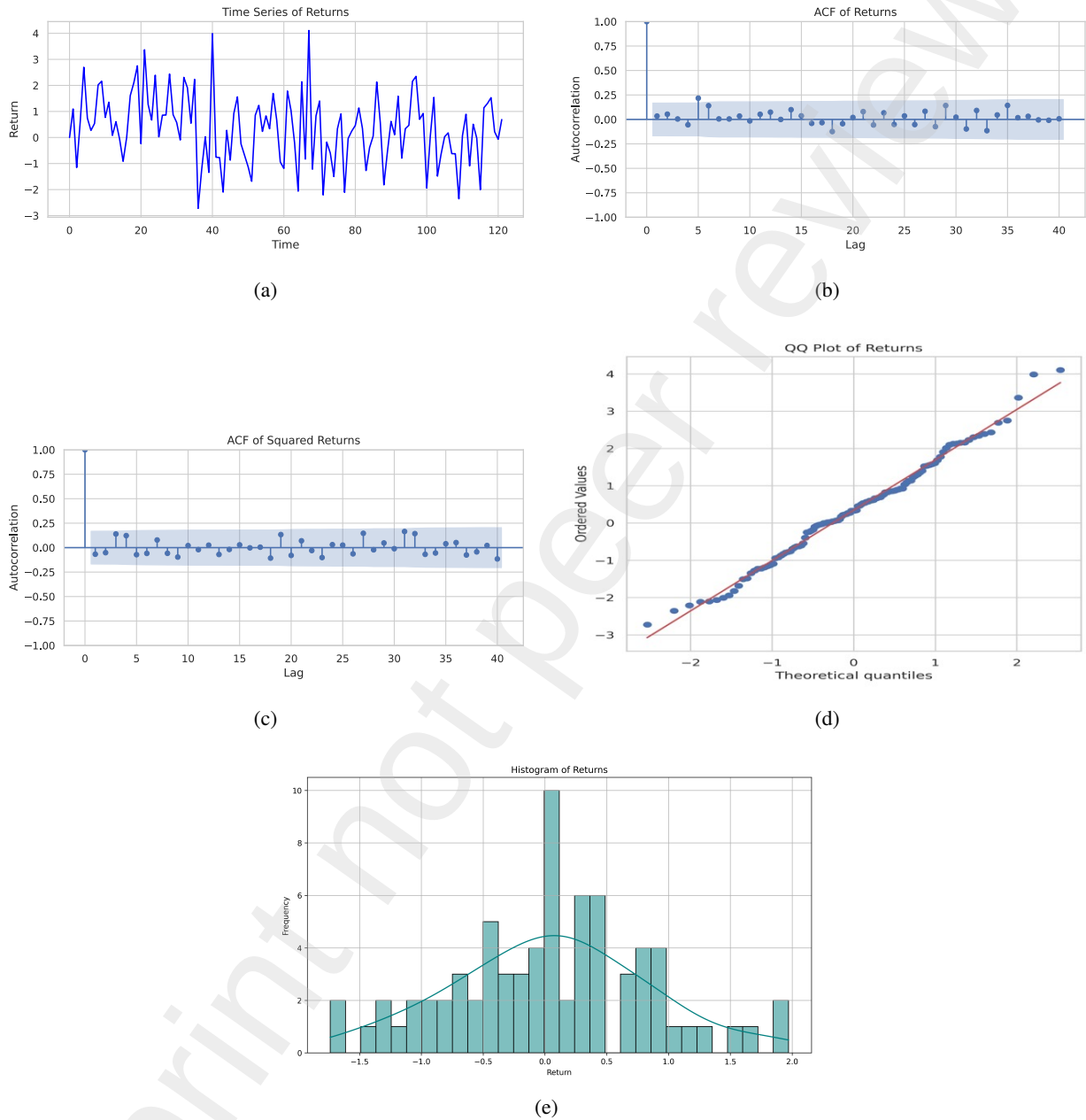


Figure 6: Return time series (a), autocorrelation function of returns (b), autocorrelation function of squared returns (c), QQ-plot of returns and (d) histogram of return distribution when percentage of agents with high financial literacy is 10% and percentage of agents with high IQ is 10% for 1 simulation run

6. Conclusion

Given that influencing the cognition of investors is difficult, increasing financial literacy of investors can influence their decision-making in times of rumor spread and decrease susceptibility to rumors. Our findings demonstrate that investor attributes play a critical role in shaping susceptibility to rumors and the subsequent spread. Our study provides a novel model of rumor spread by integrating investor cognition, financial literacy and social network effects (Ecker et al., 2022). To our knowledge, this study is the first to link investors' financial literacy and cognition to asset price dynamics during rumors. It evaluates a behavioral mechanism to reduce volatility during rumors. As the percentage of investors with high financial literacy and cognition increases, asset price volatility decreases. Existing agent-based models show that volatility increases if herding increases in the markets (Lux and Marchesi, 1998; Westerhoff and Reitz, 2003; Lux and Marchesi, 1999). Our model demonstrates the ability to replicate key stylized facts observed in financial markets in the event of rumors. It adds to the existing literature on interacting agents in ABMs that demonstrate the capability of generating key stylized facts of fat tails in returns and volatility clustering (Lux and Marchesi, 1998; Alfaro et al., 2008; Lux and Marchesi, 1999; LeBaron et al., 1999).

Regulators can change financial behavior of investors and reduce their impulsive decisions during rumors by employing just-in-time financial education (Carpena et al., 2019). Research on reducing the impact of misinformation suggests designing platforms that seek to reduce the exposure to misinformation (Chakraborty et al., 2024) and preemptive and reactive interventions (Ecker et al., 2022). The findings of our study encourage a shift from reactive to preventive regulation in dealing with market rumors by increasing financial literacy of investors. Regulators can proactively monitor the spread of misinformation by financial influencers and manipulative posts on social media platforms. Collaborations with technology firms to flag and down-rank misleading financial content could prevent rumors. Additionally, experimental studies can test how specific financial literacy interventions change susceptibility to rumors. Our study has important implications in terms of regulators actively monitoring social media posts and investing in financial literacy of investors which in turn can bring stability in asset prices in the face of rumors. When investors' financial decision-making is limited by their financial literacy and cognition, regulators can nudge investors to make less risky investments, providing a choice architecture which can mitigate the effect of rumors on asset prices (Thaler and Sunstein, 2021).

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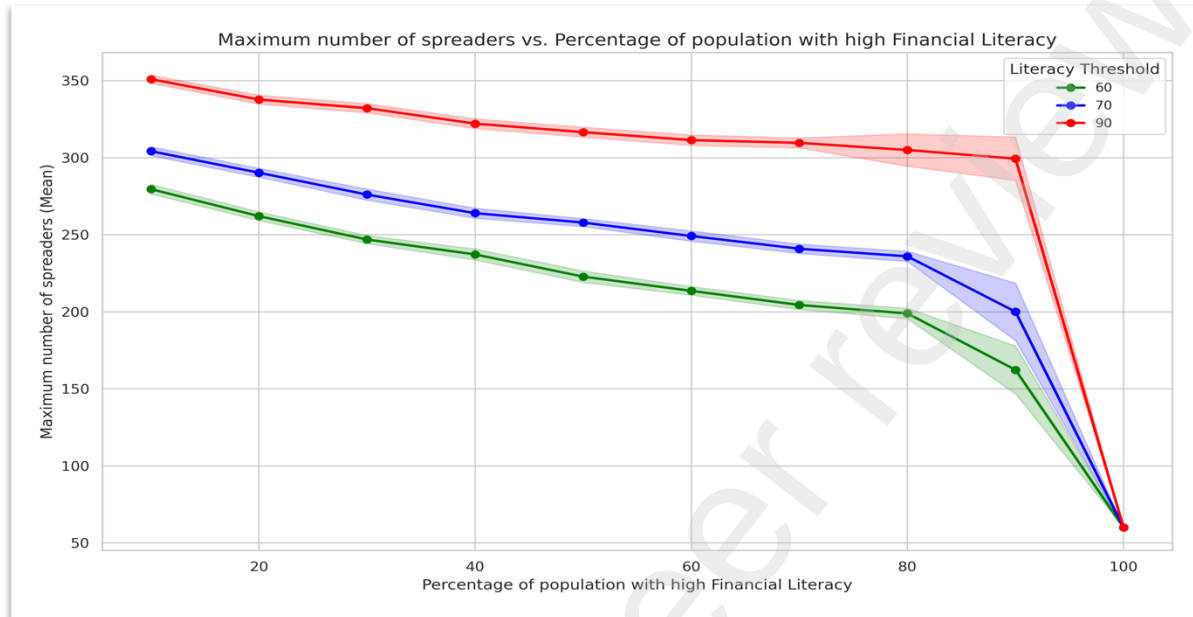
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An Agent-Based Model of Rumor-Induced Volatility in Financial Markets

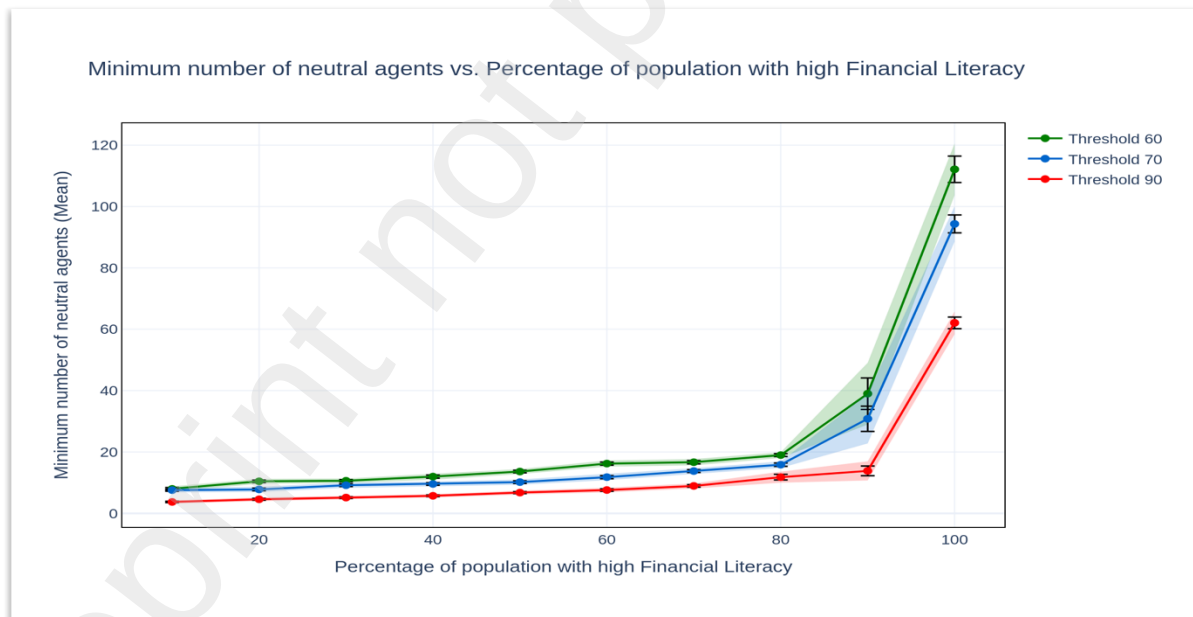
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Appendix 1

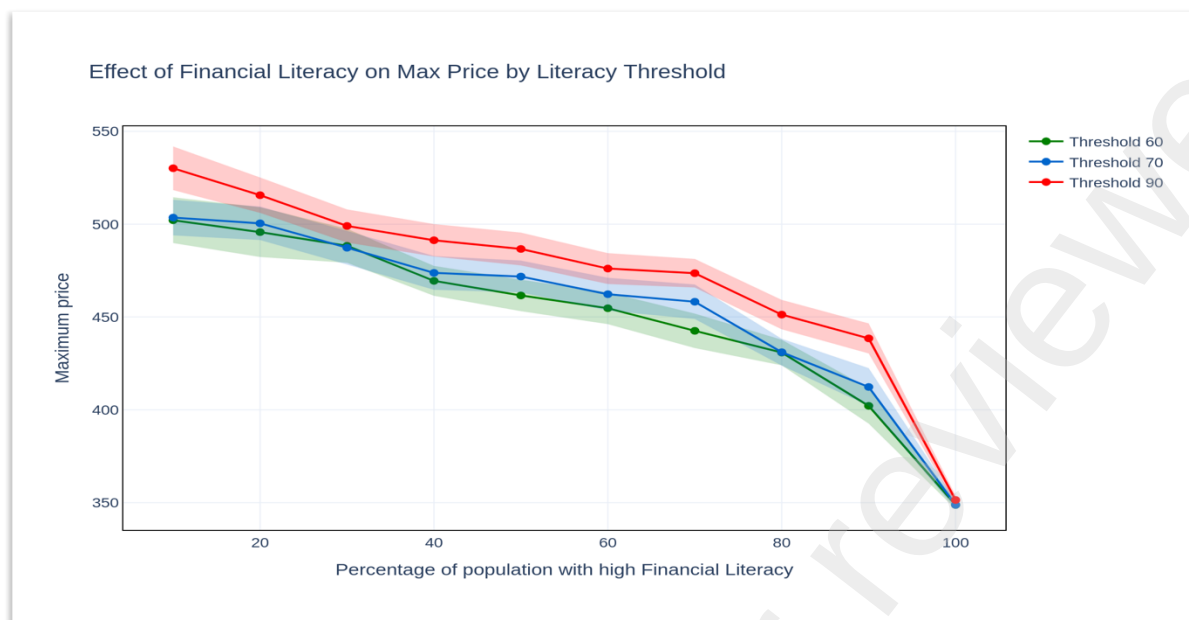
Sensitivity analysis of all parameters on the experiment when percentage of agents with high financial literacy is varied from 10% to 100% and percentage of agents with high IQ is fixed at 10% with rumor type “positive” and educate ‘ON’ with all other parameters values as in Table 1.



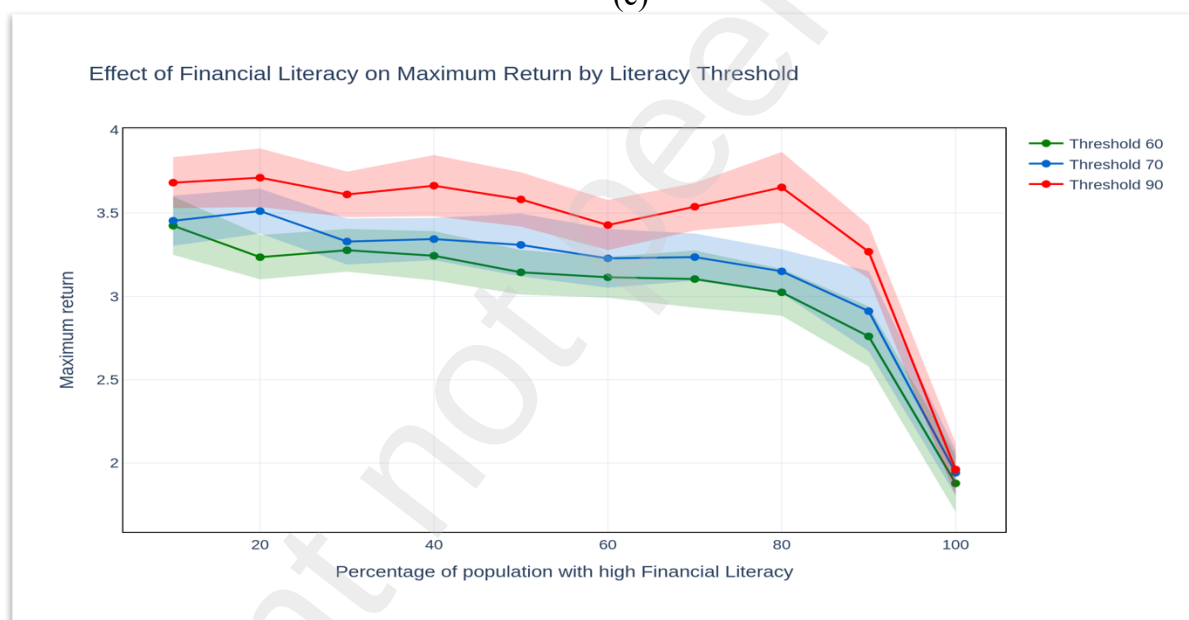
(a)



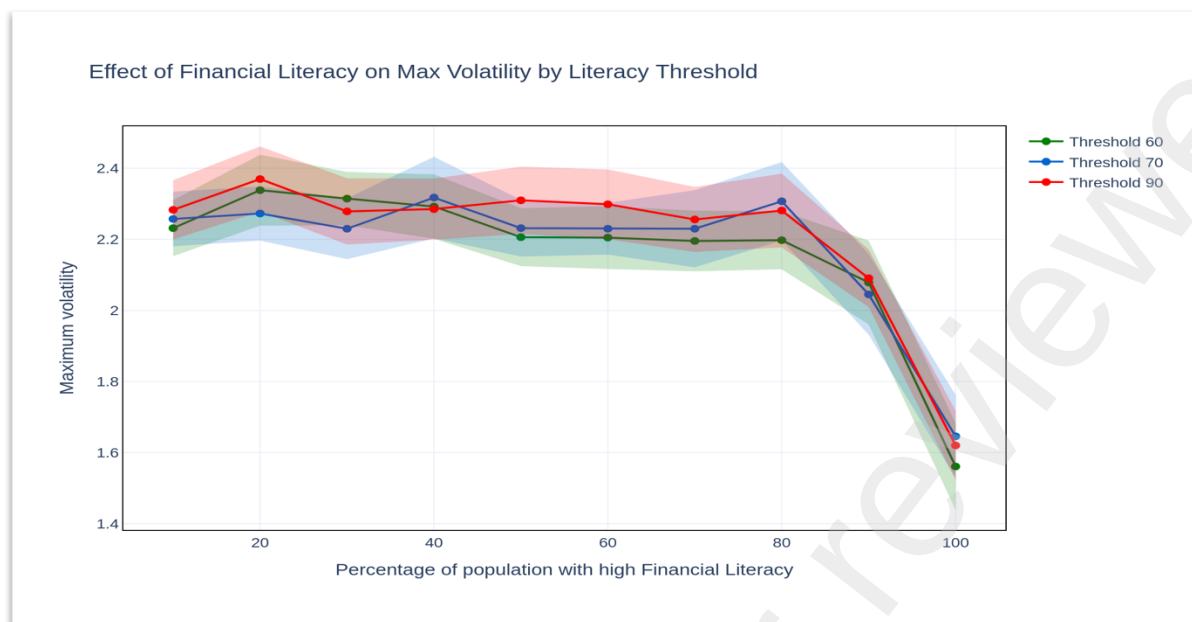
(b)



(c)



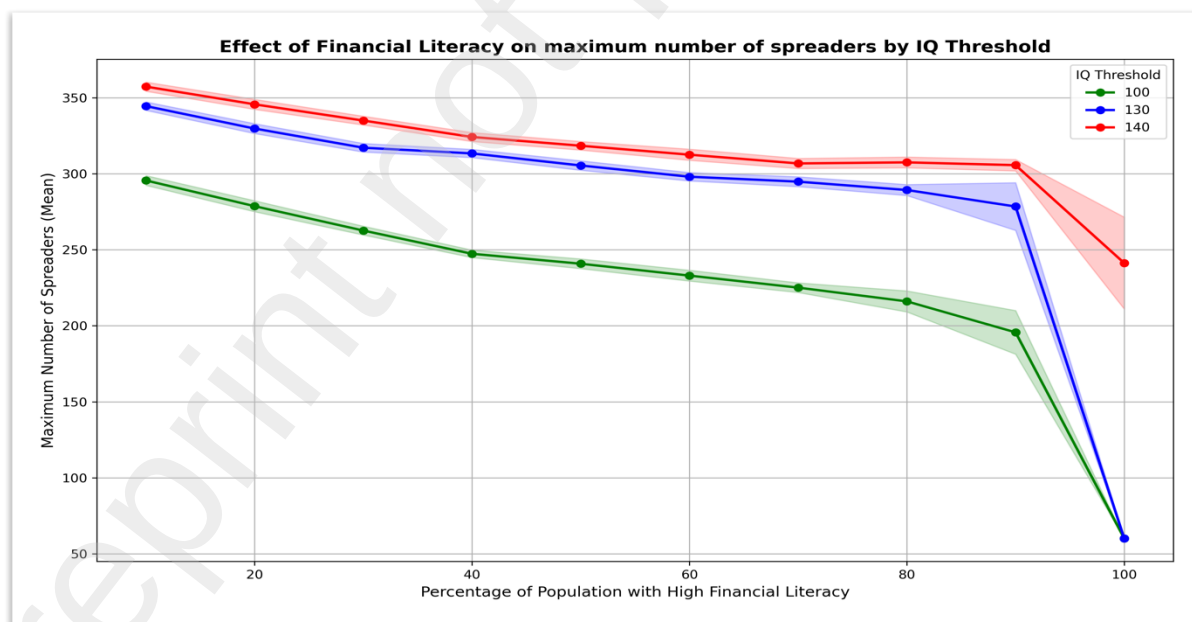
(d)



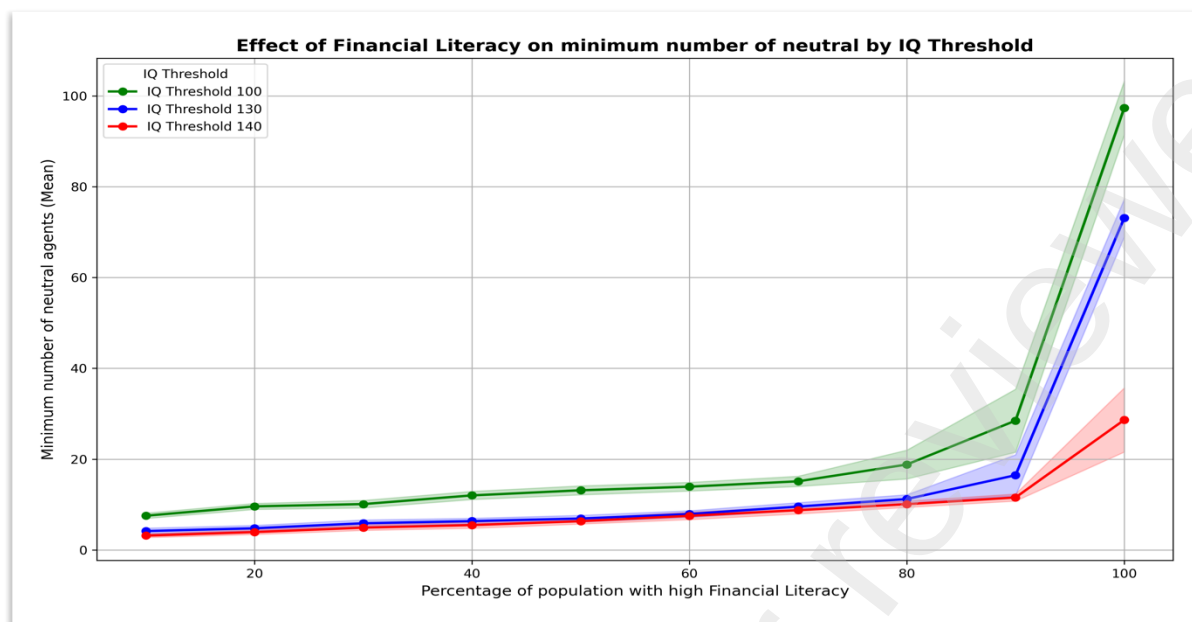
(e)

Figure1: Effect of different literacyThreshold on maximum number of spreaders (a), minimum number of neutral (b), maximum price (c), maximum return (d), maximum volatility (e) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate

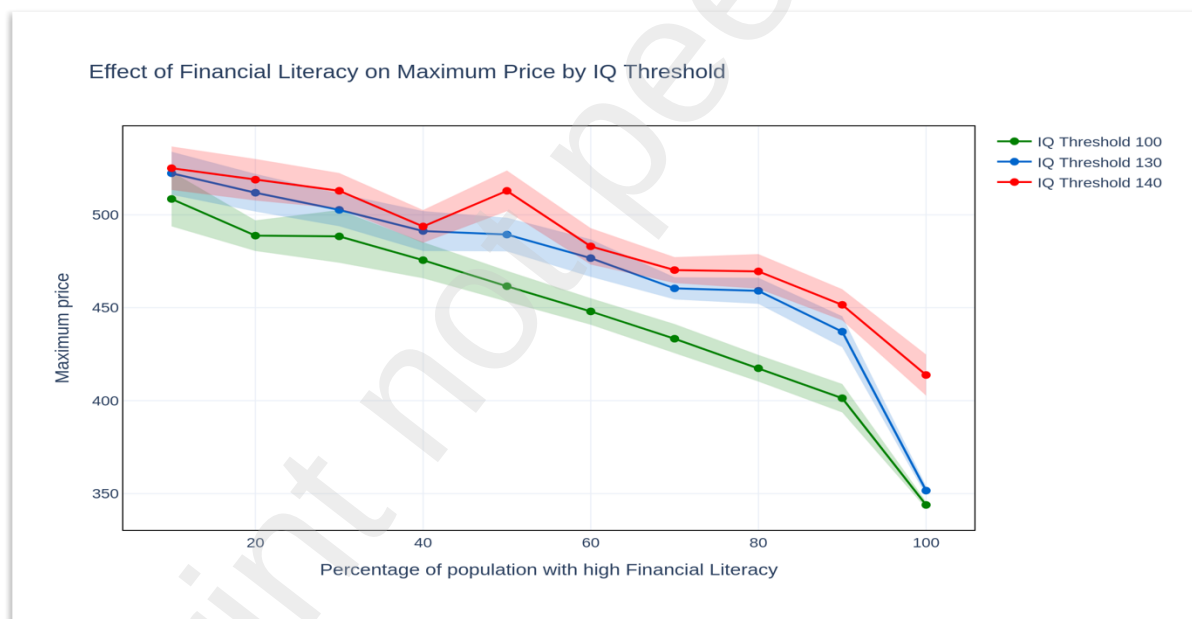
IQ Threshold



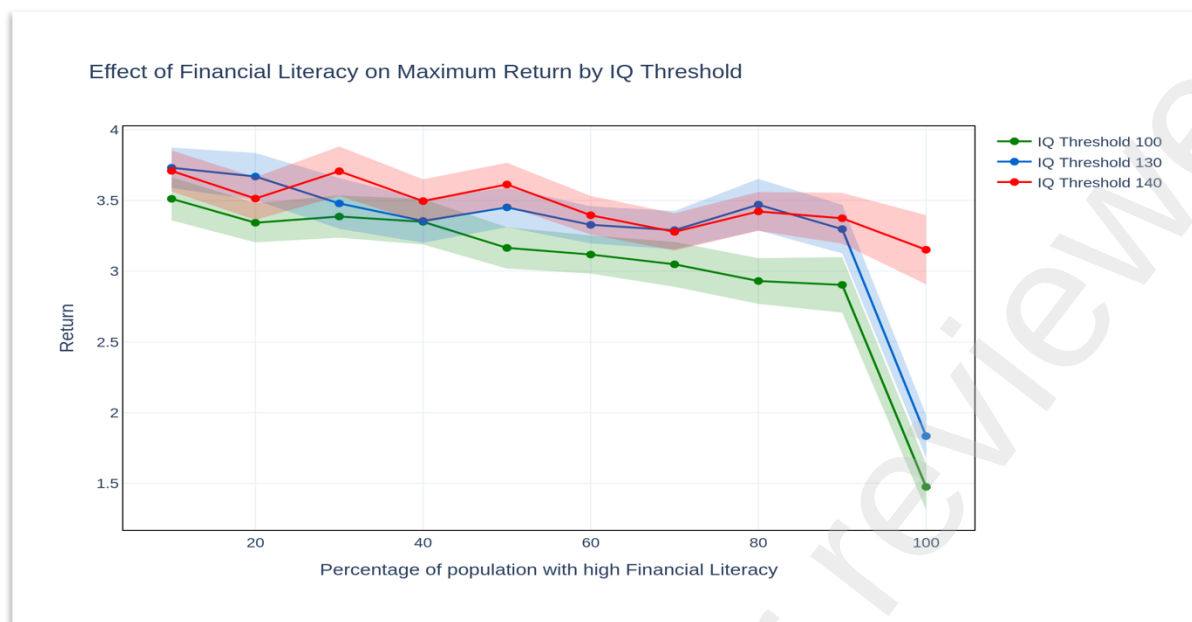
(a)



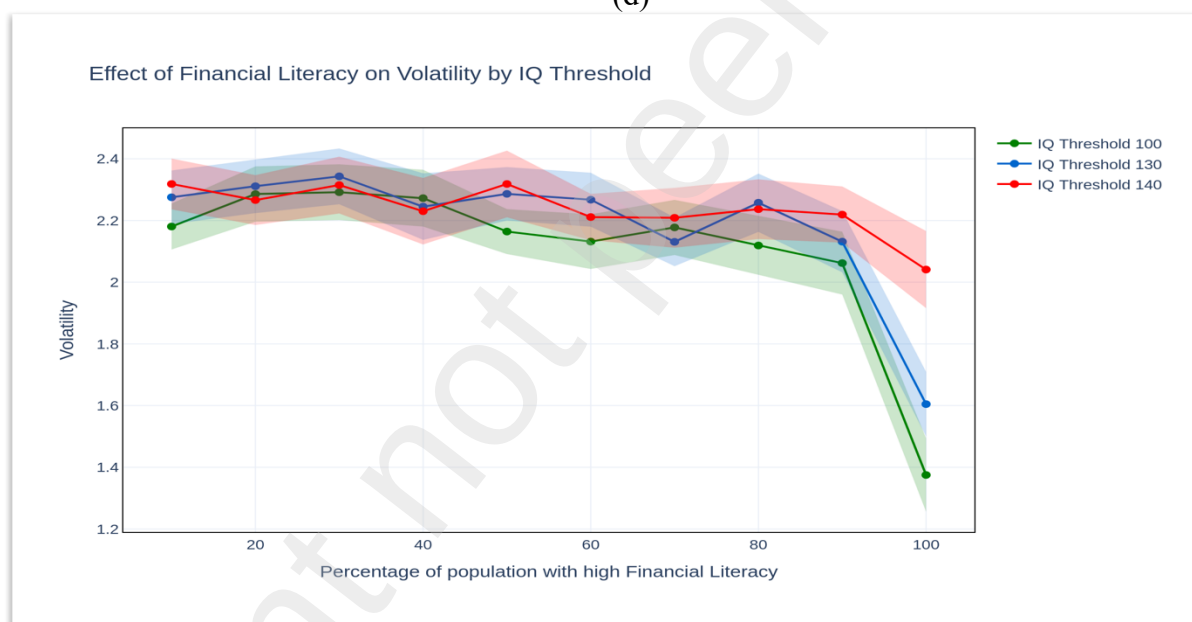
(b)



(c)



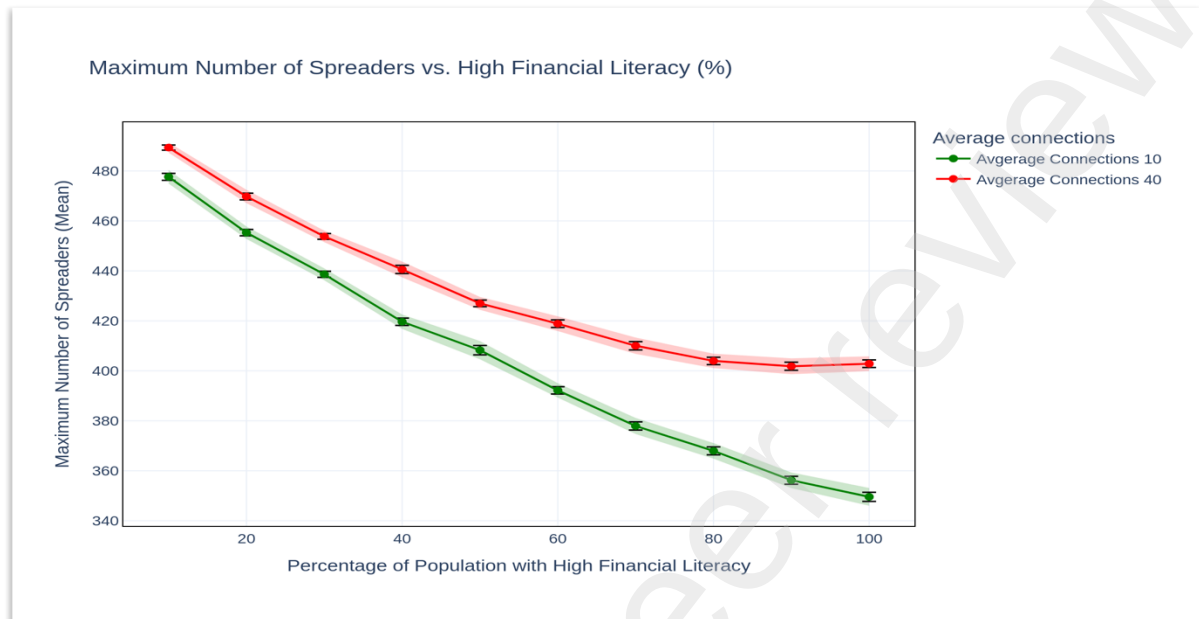
(d)



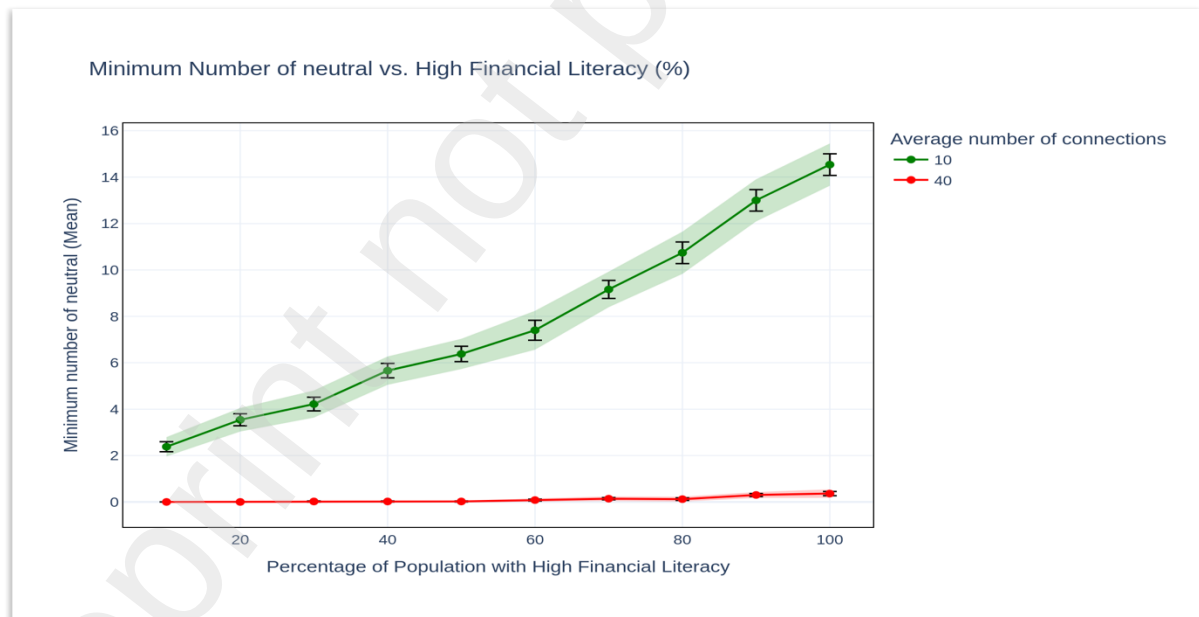
(e)

Figure 2: Effect of different iqThreshold on maximum number of spreaders (a), minimum number of neutral (b), maximum price (c), maximum return (d), maximum volatility (e) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate

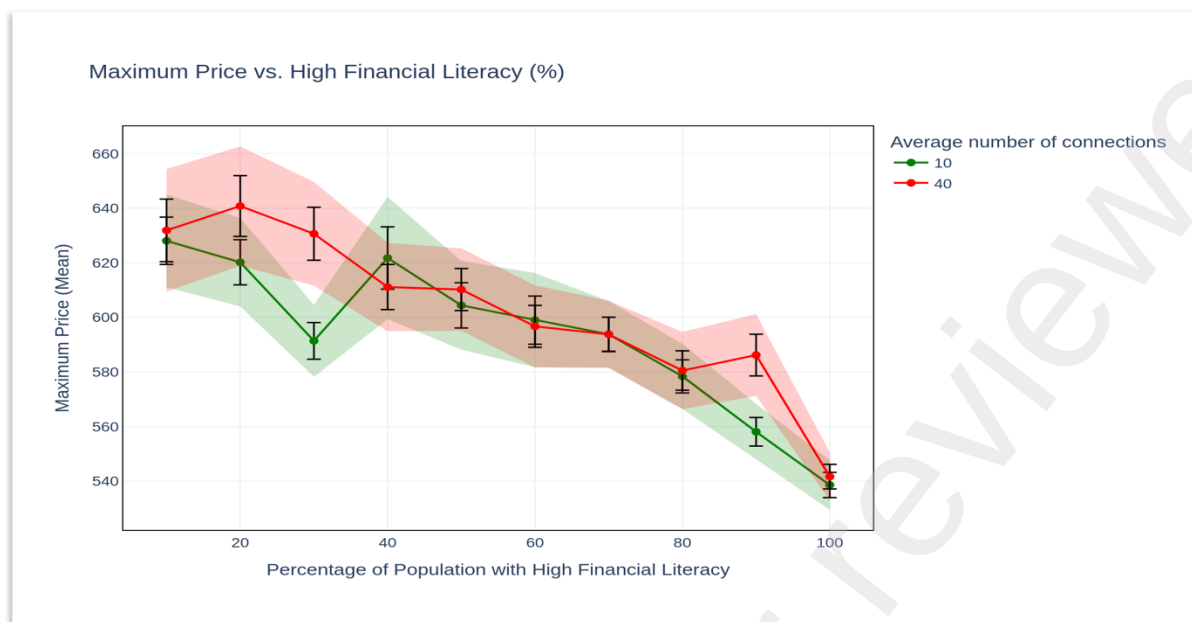
Average Connections



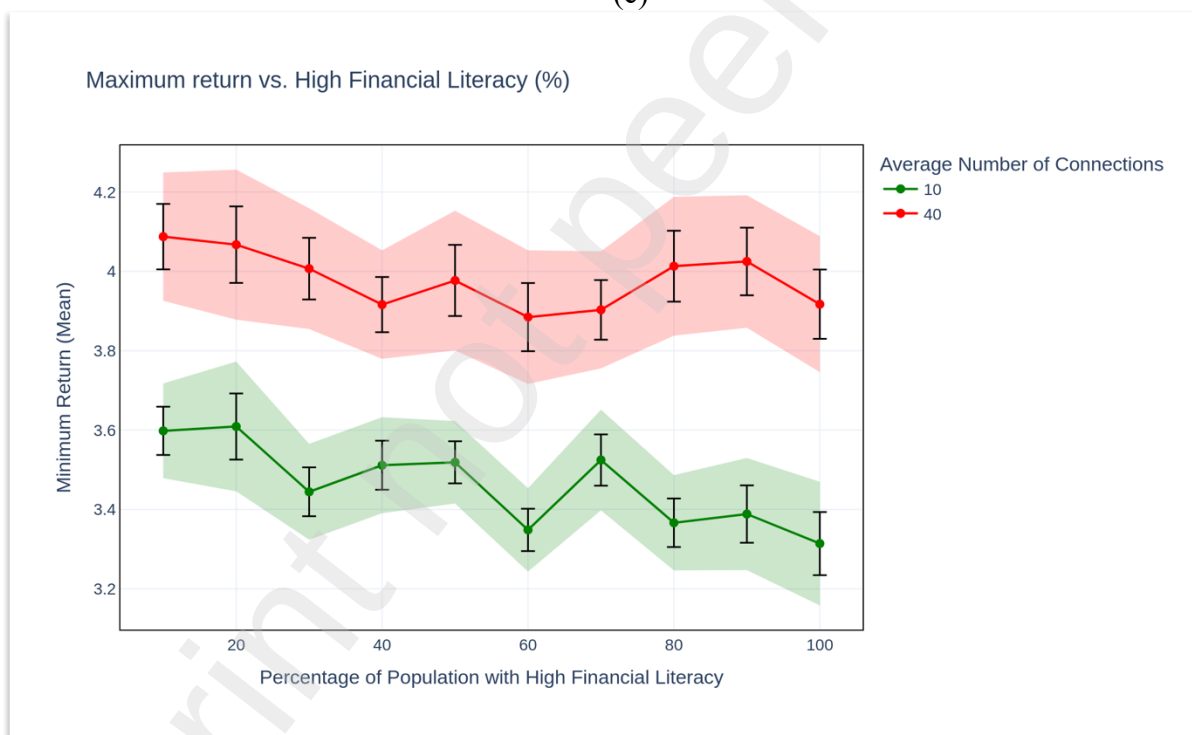
(a)



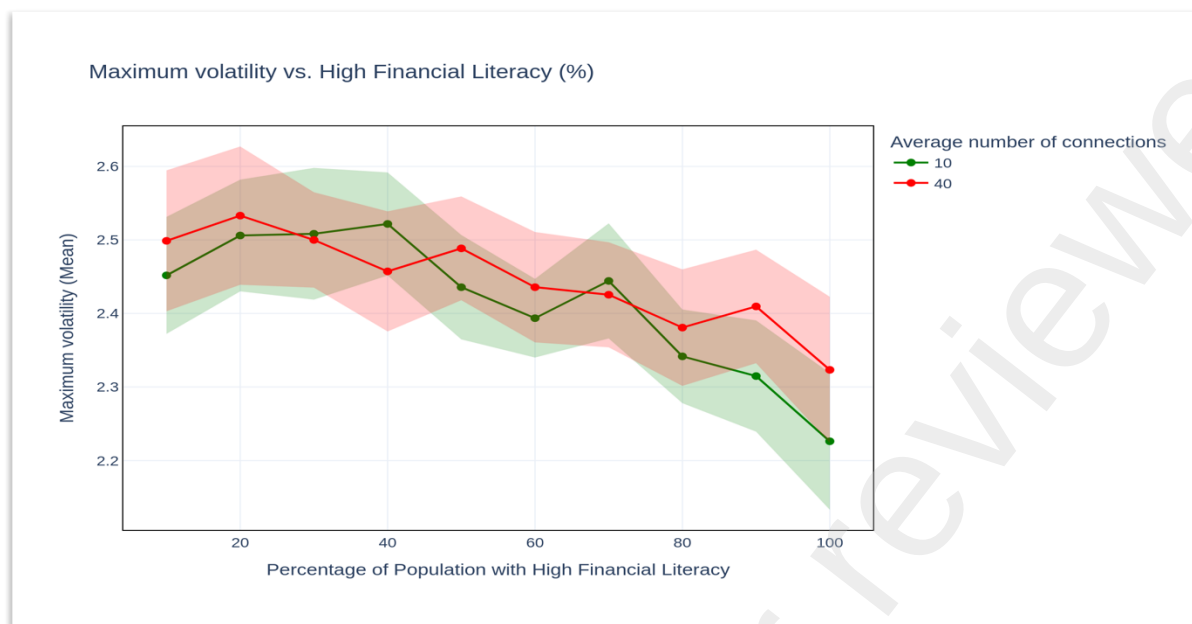
(b)



(c)



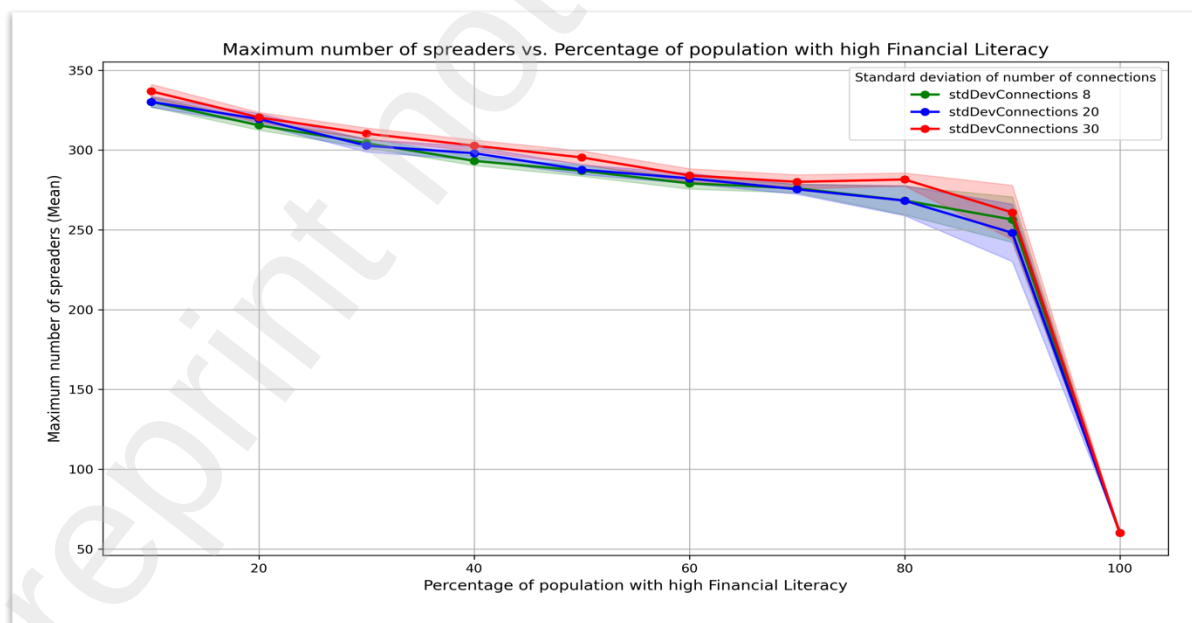
(d)



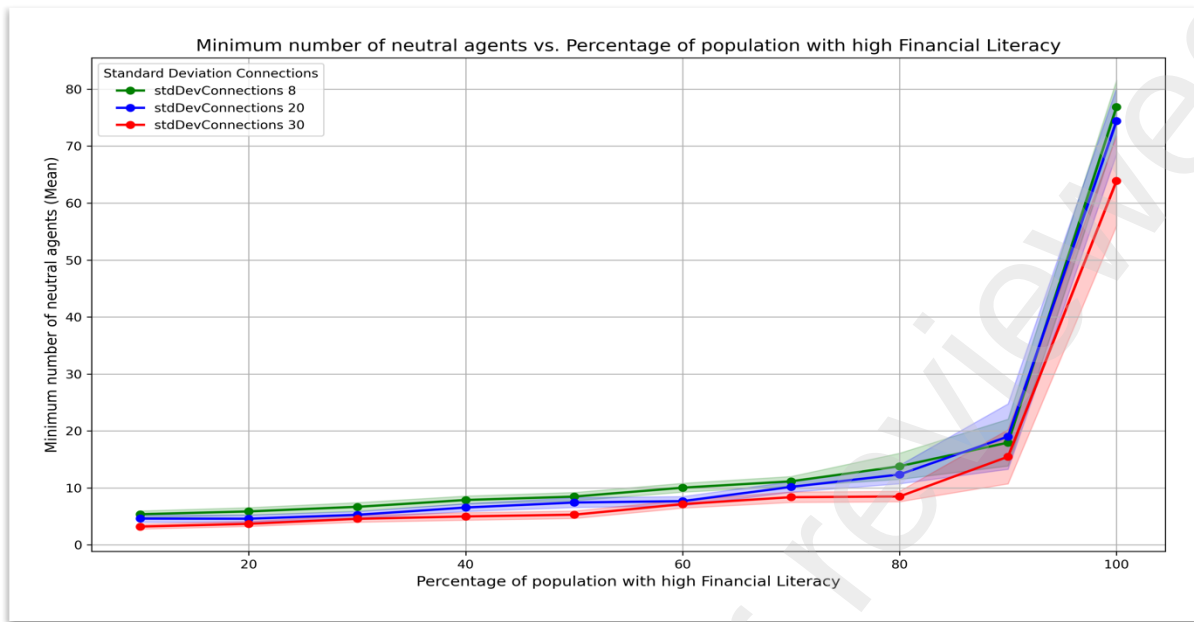
(e)

Figure3: Effect of different averageConnections on maximum number of spreaders (a), minimum number of neutral (b), maximum price (c), maximum return (d), maximum volatility (e) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate

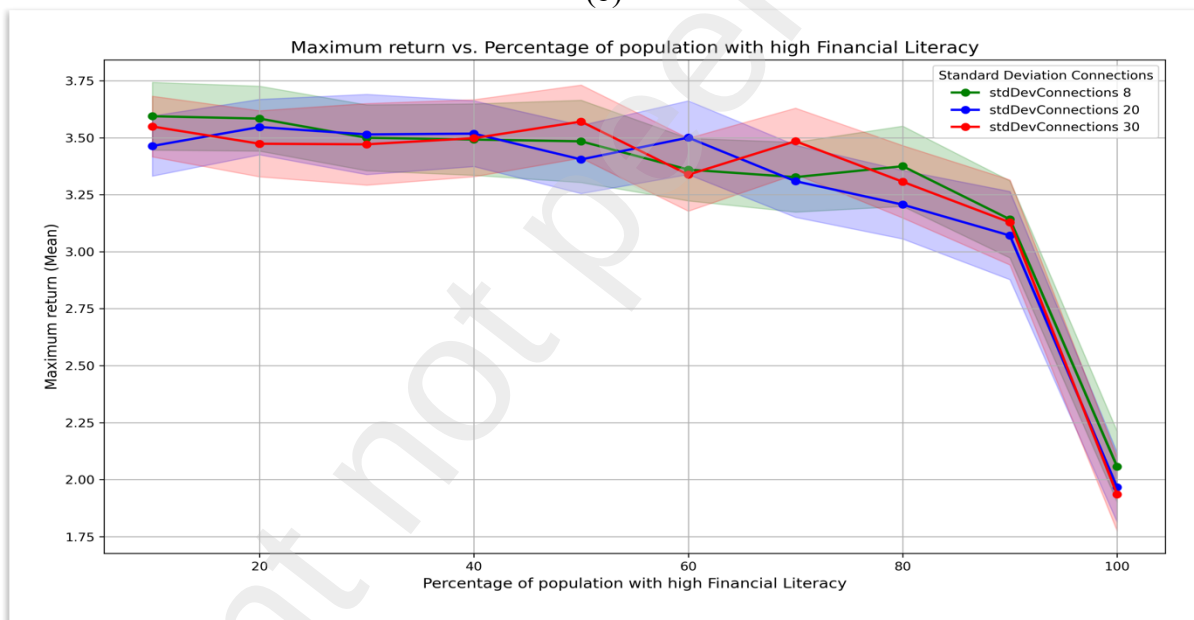
Standard Deviation connections



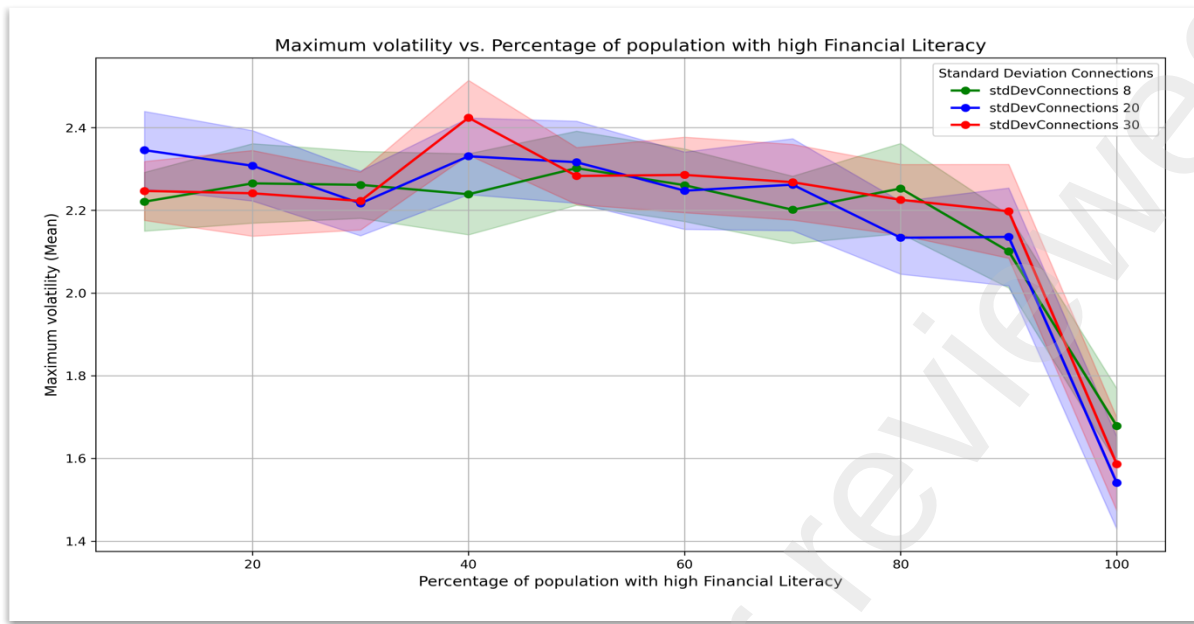
(a)



(b)



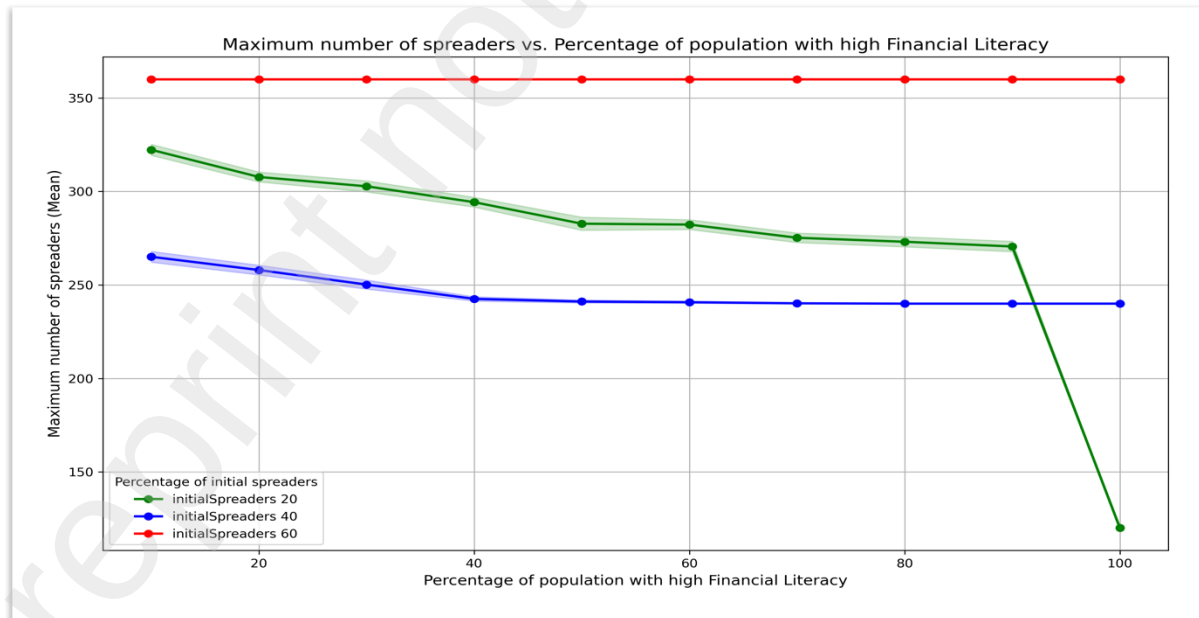
(c)



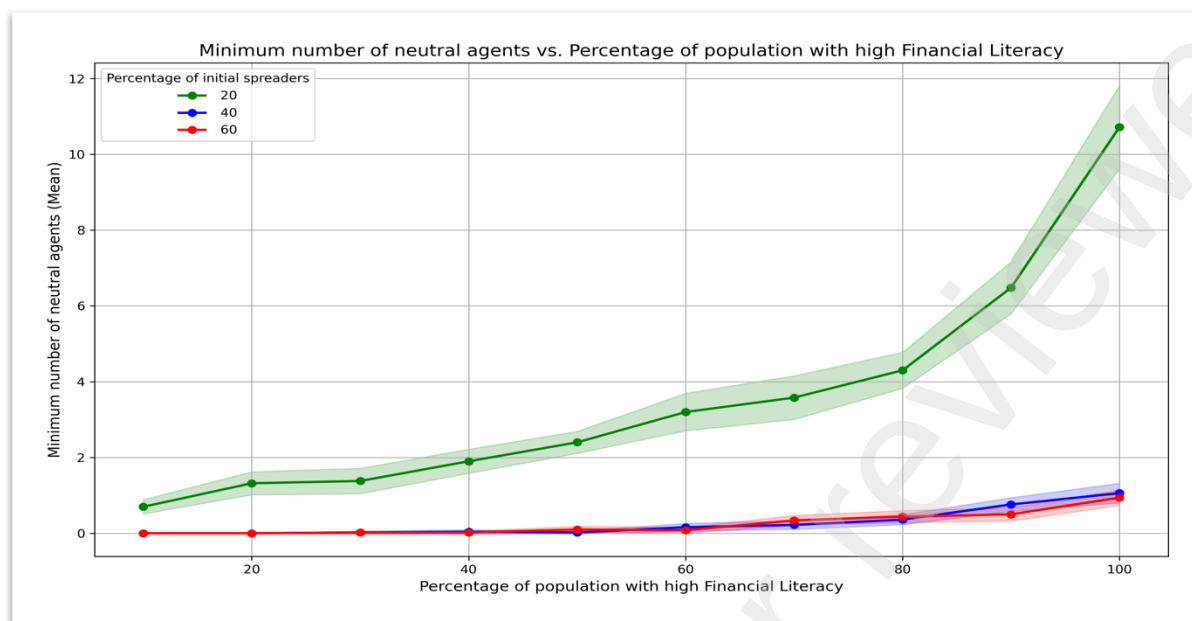
(d)

Figure 4: Effect of different stdDevConnections parameter on maximum number of spreaders (a), minimum number of neutral (b), maximum return (c), maximum volatility (d) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate

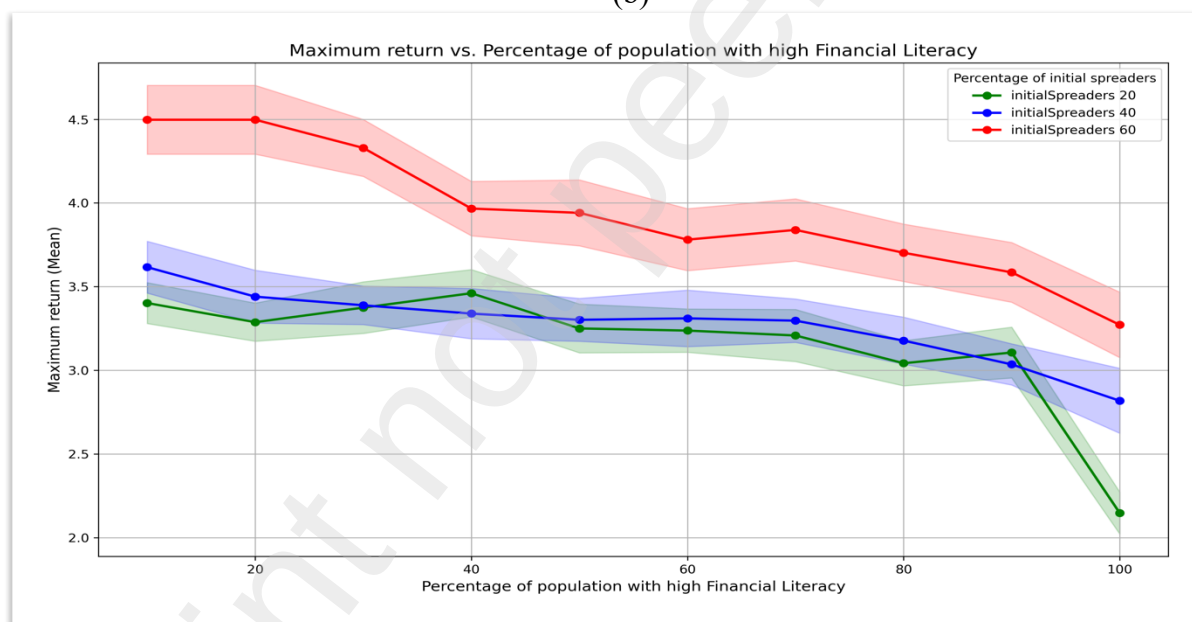
Initial Spreaders



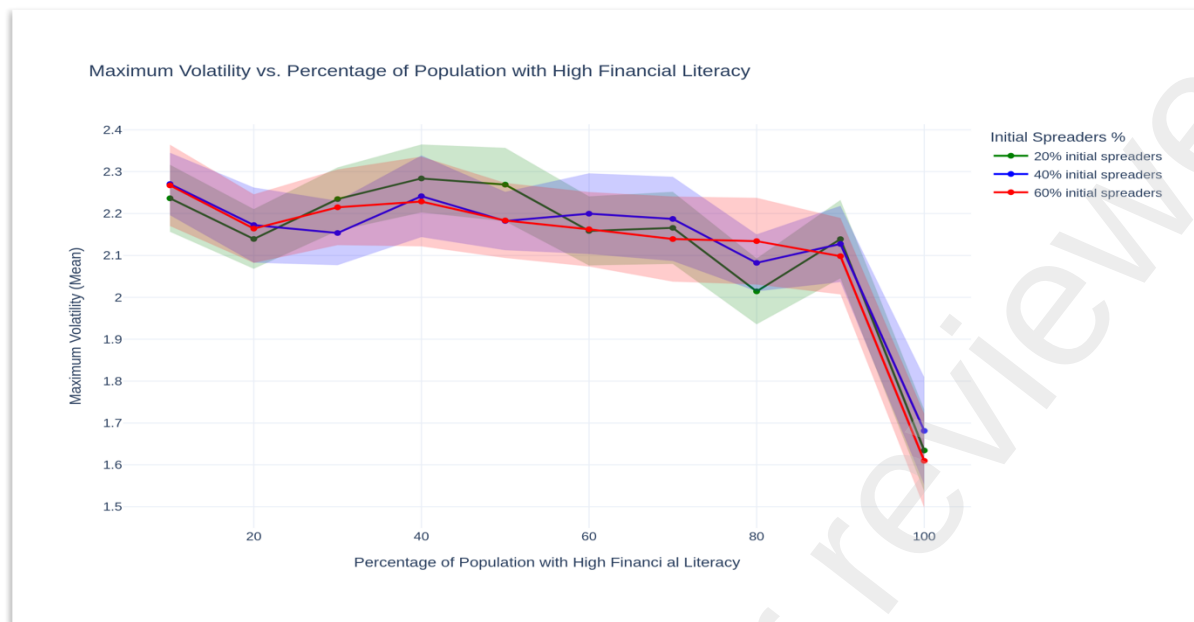
(a)



(b)



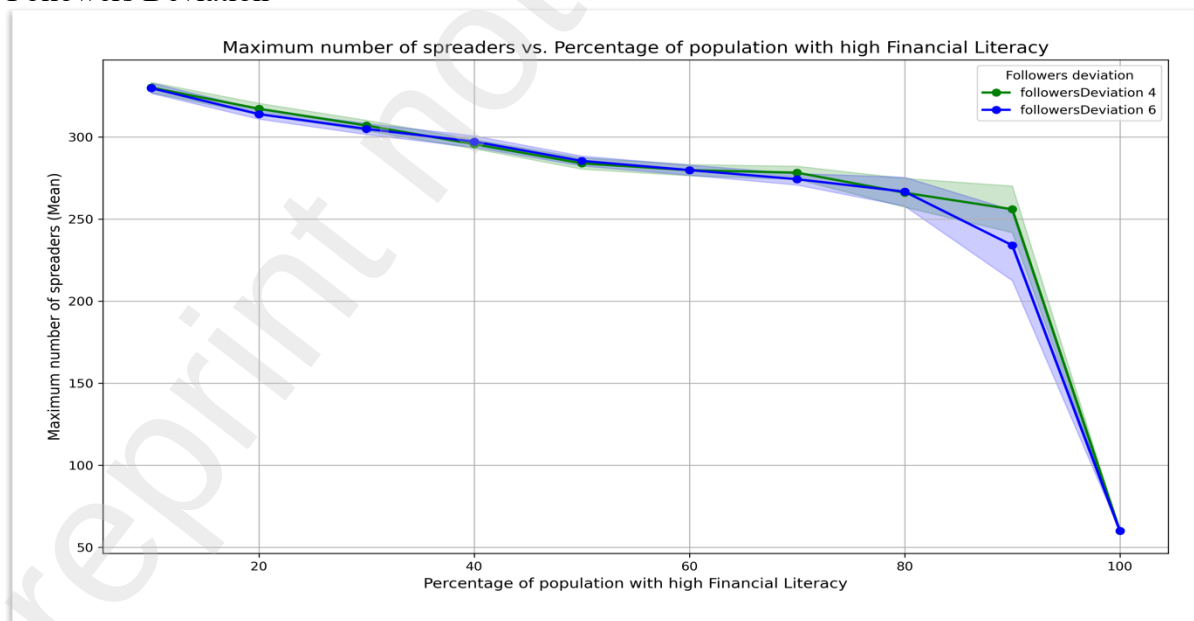
(c)



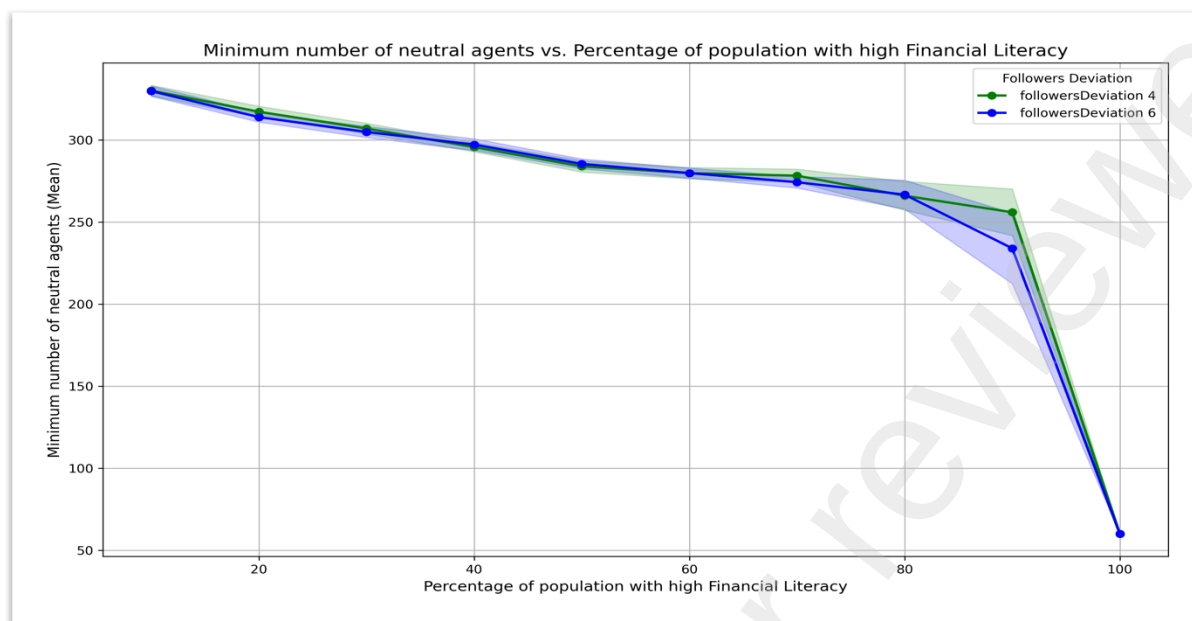
(d)

Figure 5: Effect of different percentage of initial spreaders parameter on the maximum number of spreaders (a), minimum number of neutral (b), maximum return (c) and maximum volatility (d) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate

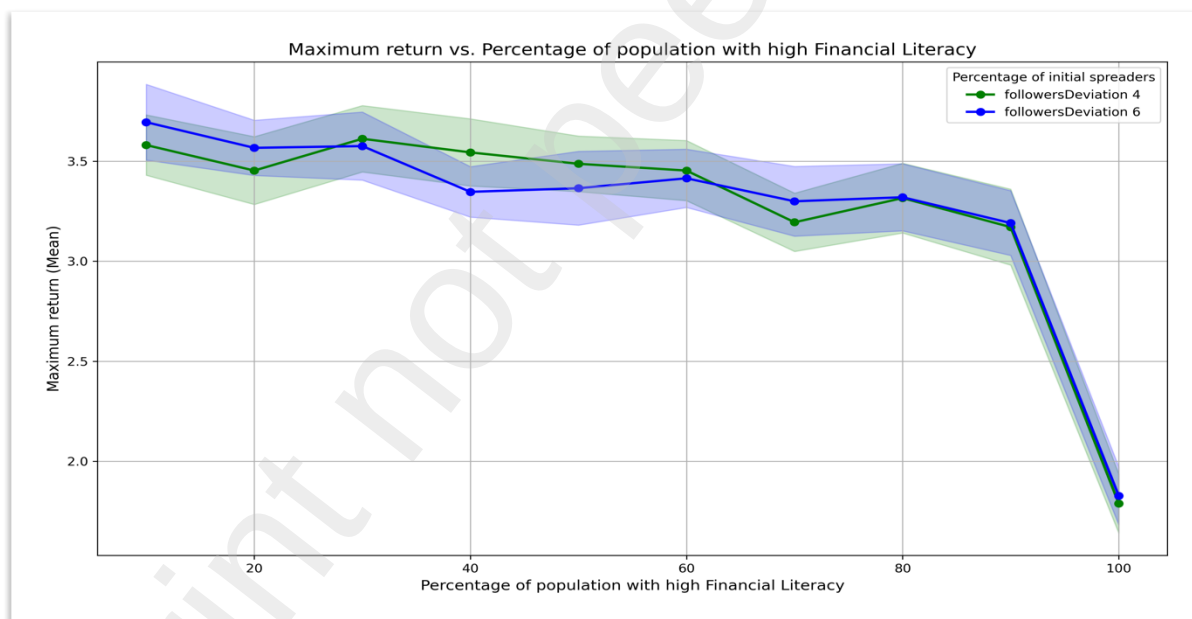
Followers Deviation



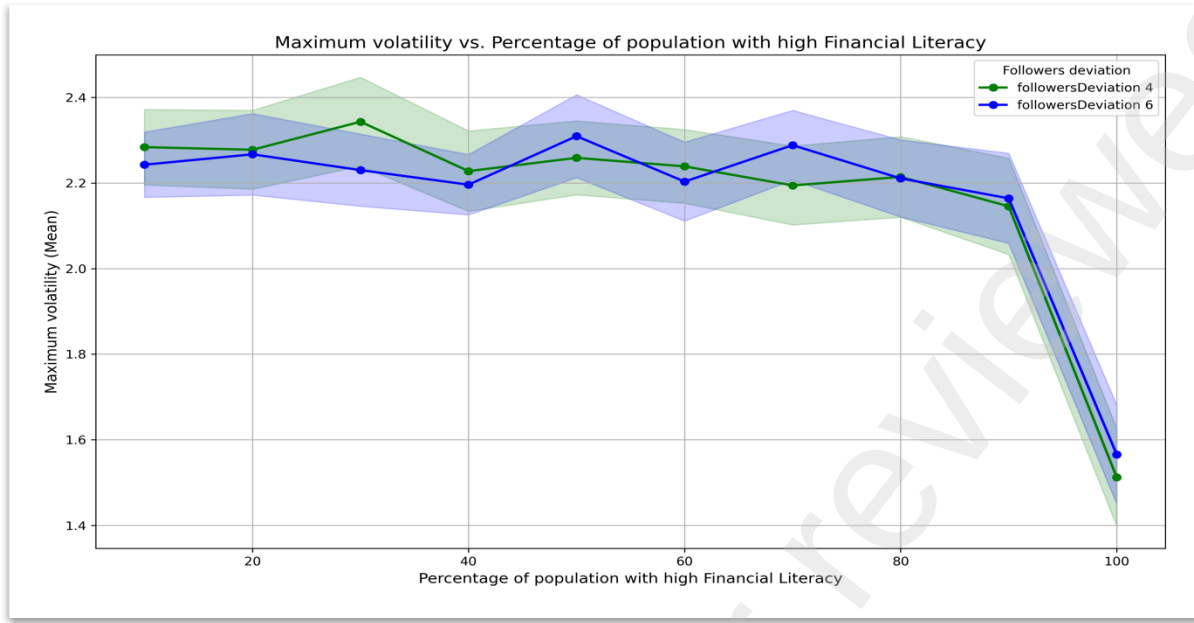
(a)



(b)



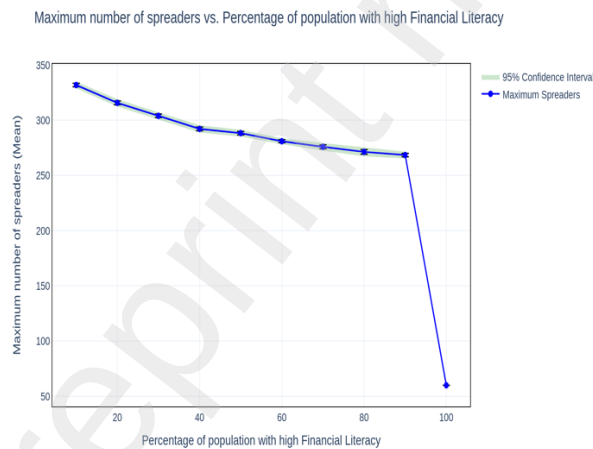
(c)



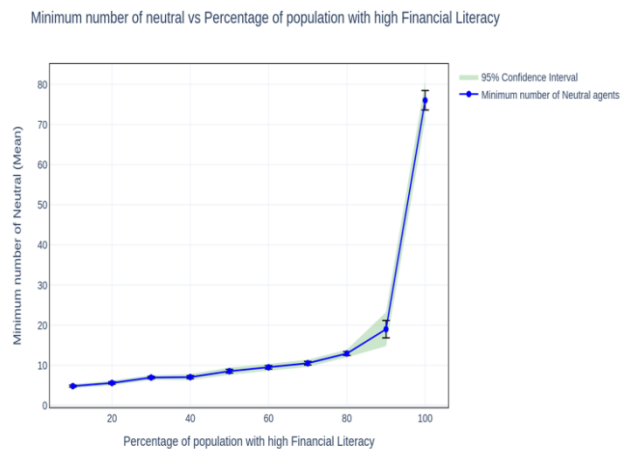
(d)

Figure 6: Effect of different followersDeviation parameter on the maximum number of spreaders (a), minimum number of neutral (b), maximum return (c) and maximum volatility (d) across 50 simulation replications. The shaded region represents 95% confidence interval of the mean estimate

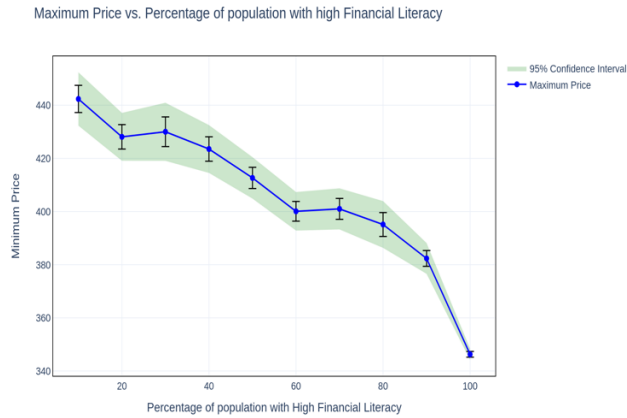
Results of the experiment when percentage of agents with high financial literacy is varied from 10% to 100% and percentage of agents with high IQ is fixed at 10% with rumor type “positive” and educate ‘ON’ with $a = 0.1$, $b = 75$ and all other parameters values as in Table 1.



(a)



(b)



(c)

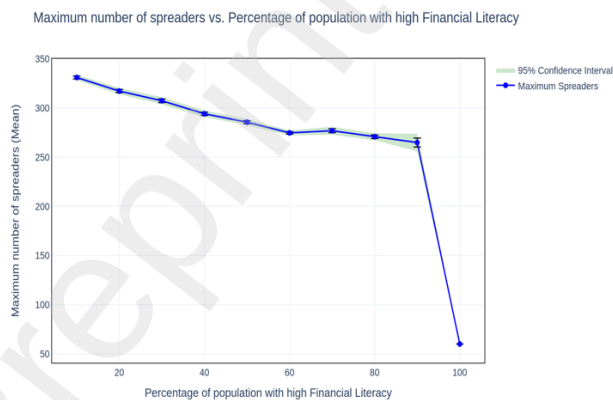


(d)



(e)

Figure 7: Percentage of agents with high *financial literacy* vs maximum agents in *spreader* state at any point in the simulation (a), minimum number of agents in *neutral* state (b), maximum price (c), maximum return (d) and maximum volatility (e) across 30 simulation replications. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate



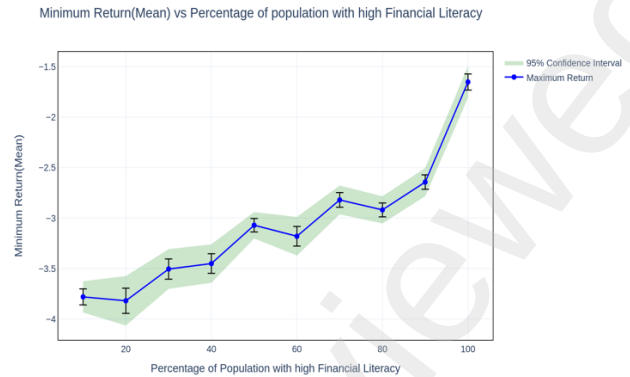
(a)



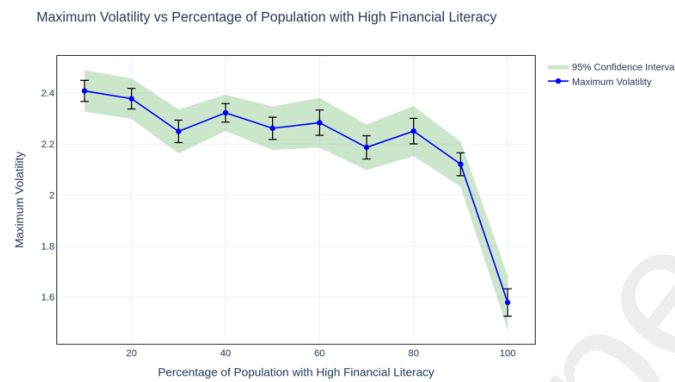
(b)



(c)



(d)

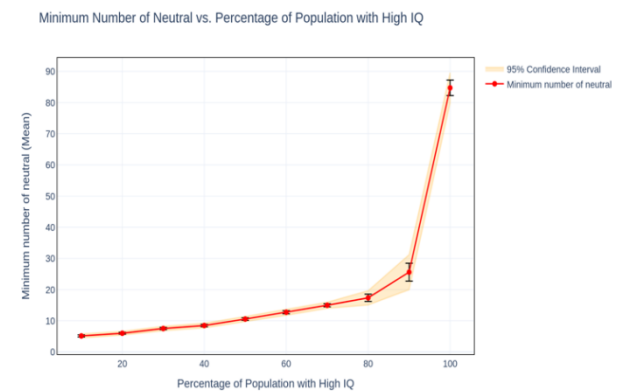


(e)

Figure 8: Percentage of agents with high *financial literacy* vs maximum agents in *spreader* state at any point in the simulation (a), minimum number of agents in *neutral* state (b), minimum price (c), minimum return (d) and maximum volatility (e) across 30 simulation replications for a negative rumor. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate



(a)



(b)

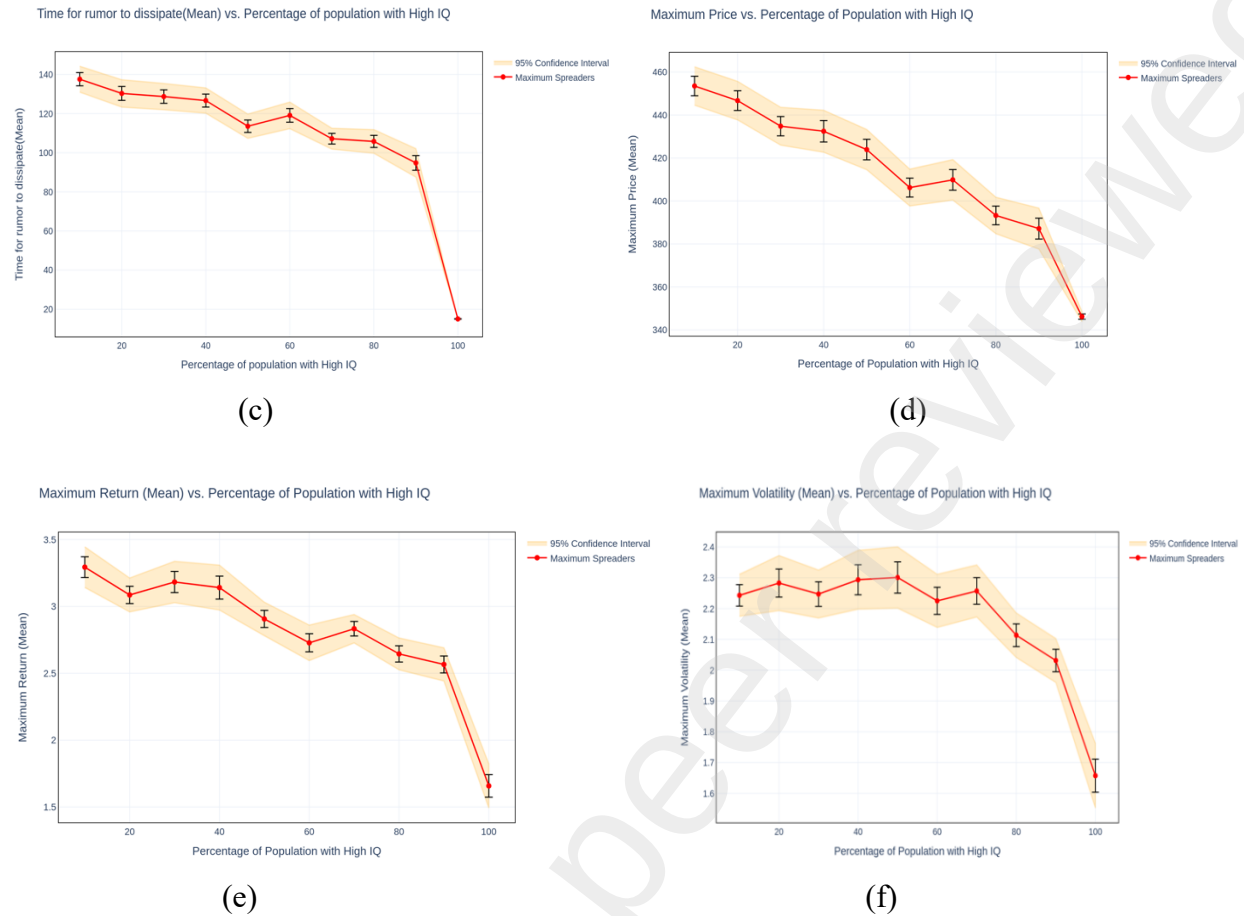
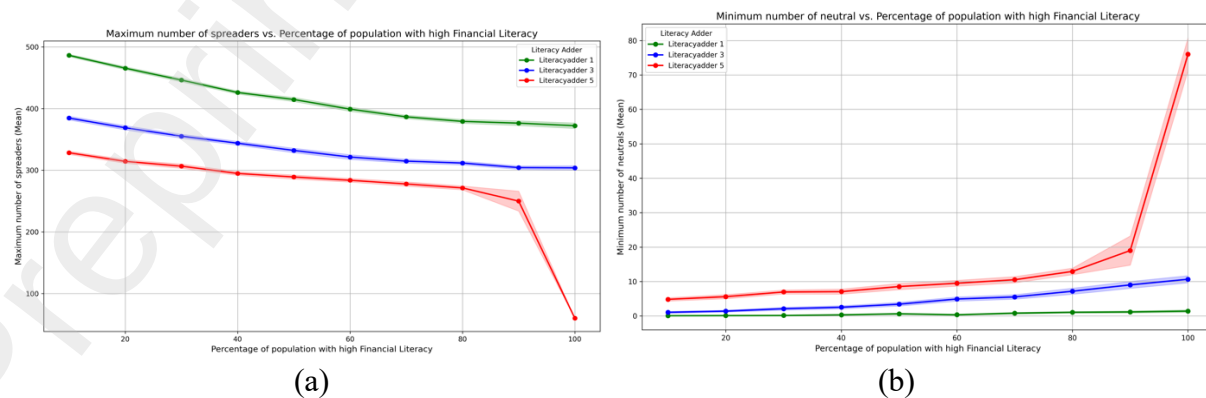
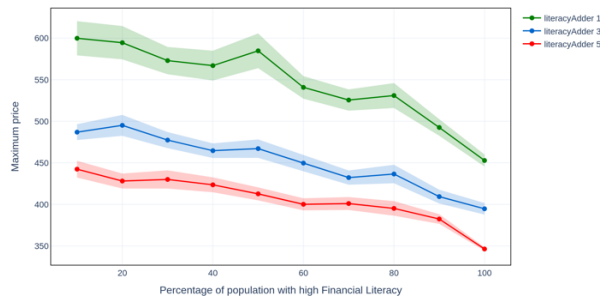


Figure 9: Percentage of agents with high IQ vs maximum agents in *spreader* state at any point in the simulation (a), minimum number of agents in *neutral* state (b), maximum price (c), maximum return (d) and maximum volatility (e) across 30 simulation replications. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate.

Effect of Literacy Adder

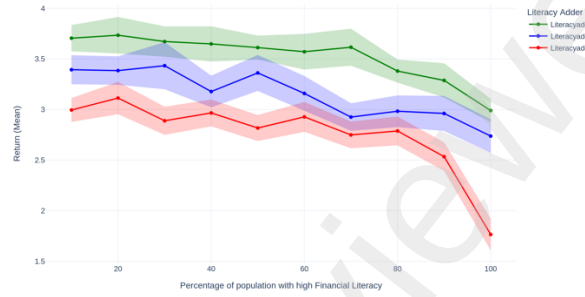


Effect of Financial Literacy on Max Price by Literacy adder



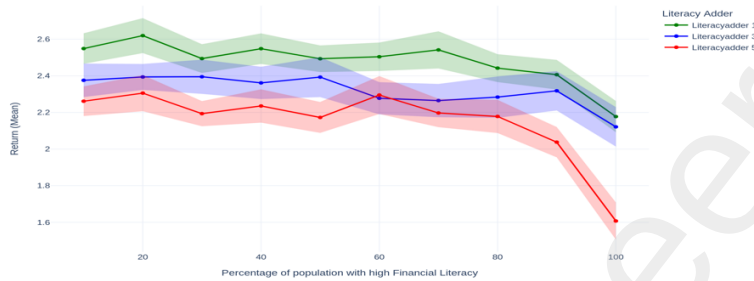
(c)

Maximum return vs. Percentage of population with high Financial Literacy



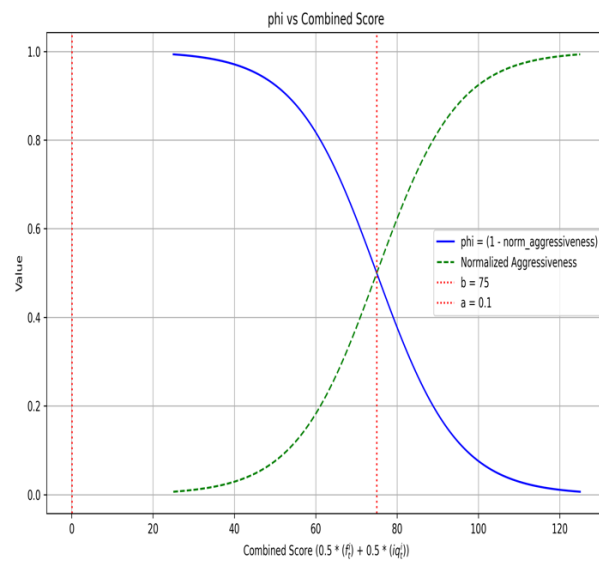
(d)

Maximum return vs. Percentage of population with high Financial Literacy

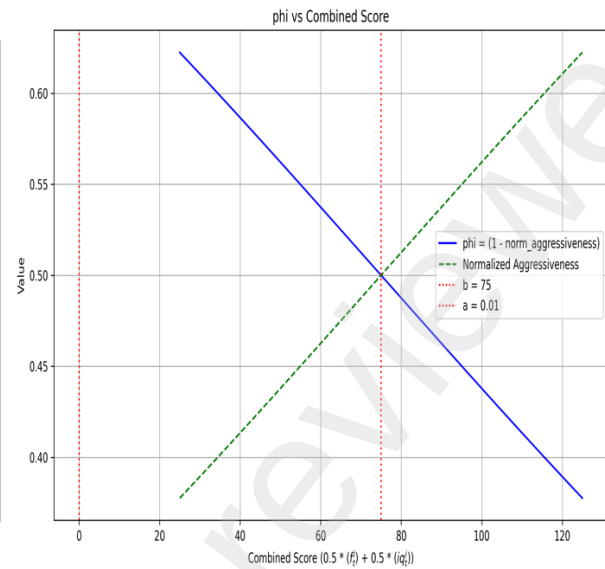


(e)

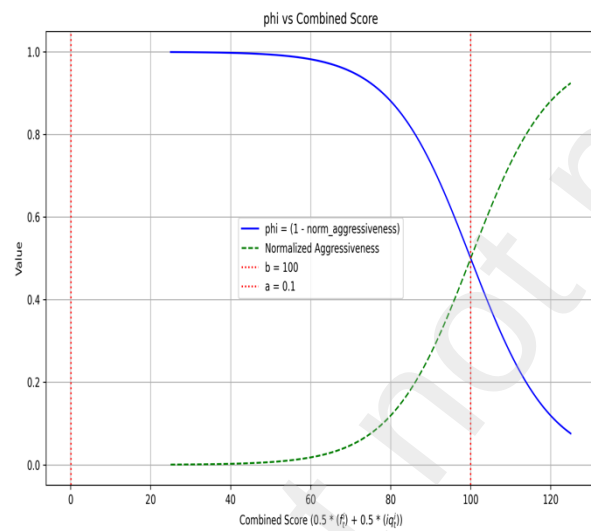
Figure 10: Percentage of agents with high *financial literacy* vs maximum agents in *spreader* state at any point in the simulation (a), minimum number of agents in *neutral* state (b), maximum price (c), maximum return (d) and maximum volatility (e) across 30 simulation replications varying literacyAdder parameter. Error bars represent ± 1 standard error. The shaded region represents 95% confidence interval of the mean estimate.



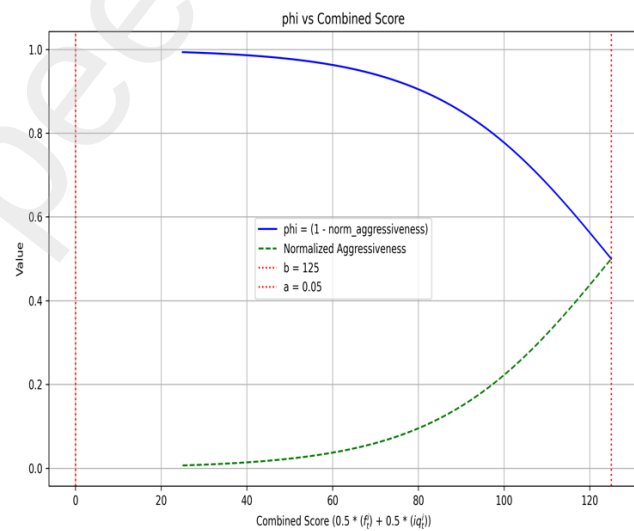
(a)



(b)



(c)



(d)

Figure 11: Effect of different values of parameters a and b on the strength of demand (ψ) of spreaders across different combined scores (c_i).

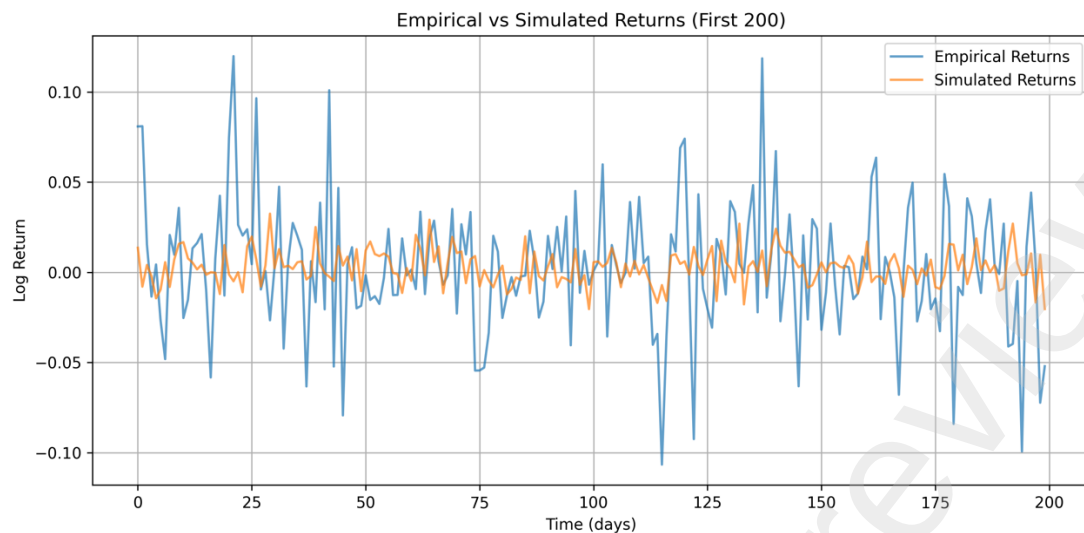


Figure 12: Empirical and simulated returns over the first 200 days using the calibrated Franke & Westerhoff (2012) model.